Embryonic-disc and extra-embryonic-membrane transcriptomes of same-length day-15 bovine conceptuses produced in vitro or in vivo: an exploratory analysis

Written by: Janaki Balasubramanian

MSc research project, Utrecht University, Department of Veterinary Medicine (supervisor: M. Belen Rabaglino)

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Abstract

This exploratory analysis compared embryonic-disc (ED) and extra-embryonic-membrane (EEM) transcriptomes of four in vitro-produced (IVP) and four in vivo-produced (IVV) day-15 bovine conceptuses of the same length, with both tissues available from each conceptus. All IVP conceptuses were inferred to be male and all IVV conceptuses female, the IVP and IVV conceptuses were carried by two recipients and two superovulated donors respectively, and an unresolved sample-sheet label marks the four IVP conceptuses. Group differences in this primary comparison can therefore not be attributed specifically to IVP. Differential-expression analysis (DESeq2, FDR < 0.05) identified 1,218 genes in the ED and 721 in the EEM, and a paired-tissue model identified 417 genes for which the IVP–IVV contrast differed between tissues. In the ED, glycosylation and lipid-metabolism gene sets were more highly expressed in the IVP group and kept this direction across within-cohort sensitivity analyses, including omission of any single conceptus; only the disc-score-adjusted version depended on one conceptus. Developmental gene sets (morphogenesis, Wnt, BMP/TGF-β and Notch) were lower in the unadjusted IVP ED comparison, but this contrast was sensitive to adjustment for an internally derived disc-enrichment score; this is compatible with differences in tissue composition or in disc-cell state, which bulk RNA-seq cannot distinguish. In the EEM, adhesion and extracellular-matrix gene sets were higher in the IVP group and kept their direction in alternative analyses, whereas lower translation and respiration gene sets reversed. X-linked genes were over-represented among DEGs, but removing them did not change the gene-set results. These findings identify candidate expression differences in a selected, confounded cohort rather than effects of IVP, and should be tested in sex- and dam-balanced designs.

Layman's summary

Food security is an important concern for a growing population, and livestock are critical for milk and meat production. In cattle breeding, embryos produced in the laboratory (in vitro production, IVP) are now used more often than embryos produced inside the cow (in vivo, IVV): in 2020, about two thirds of the roughly 1.1 million bovine embryos transferred worldwide were IVP (Viana, 2021). IVP accelerates genetic gain, but IVP embryos are lost more often during early pregnancy, mostly while the embryo elongates between day 7 and day 21. At about day 15 the embryo has two distinct parts: the embryonic disc (ED), which forms the fetus, and the extra-embryonic membranes (EEM), which form the placenta.

This project compared gene activity in the ED and EEM of IVP and IVV embryos of the same length. The two groups differed in the activity of hundreds of genes in both parts. In the embryonic disc, the IVP group showed higher activity of genes for sugar attachment to proteins (glycosylation) and fat metabolism. In the membranes, genes for cell attachment and the surrounding matrix were more active in the IVP group. Disc samples from the IVP group also had lower activity of genes typical of the disc; this could mean that the samples contained less disc tissue, that the disc cells were in a different state, or both. Because all IVP embryos in this comparison happened to be male and all IVV embryos female, and each group came from only two cows, these differences cannot be attributed to the laboratory procedure itself and need to be confirmed in larger, better-balanced studies. Understanding these differences can help improve culture conditions for laboratory-produced embryos.

Keywords

Assisted reproductive technologies
In vitro embryo production
Transcriptomics
Pregnancy
Conceptus elongation

Abbreviations

DEG	Differentially expressed gene	IVP	In vitro produced/production
ED	Embryonic disc	IVV	In vivo produced/production
EEM	Extra-embryonic membranes	LFC	Log2 fold change
FDR	False discovery rate (Benjamini–Hochberg)	LOEO	Leave-one-embryo-out
GO	Gene Ontology	NES	Normalised enrichment score
GSEA	Gene set enrichment analysis	ORA	Over-representation analysis
IFNT	Interferon tau	PCA	Principal component analysis
SVM	Support vector machine	RF	Random forest
VST	Variance-stabilising transformation	TE	Trophectoderm

Introduction

Ensuring food security in the face of a continuously growing global population is of great importance. Within the cattle industry, enhancing reproductive efficiency is a crucial necessity to guarantee a steady supply of milk and meat (Barkema et al., 2015). The successful implementation of assisted reproductive technologies in cattle has reduced generational intervals and increased genetic gain in breeding populations, with positive economic outcomes for both beef and milk production (Granleese et al., 2015; Moore & Hasler, 2017). In vitro produced (IVP) embryos are produced more each year: in 2020, two thirds of bovine embryos transferred worldwide were IVP, compared with one third produced by traditional in vivo (IVV) methods (Viana, 2021). IVP offers advantages in resource utilisation, labour efficiency and facility management. It also makes it possible to use oocytes from animals no longer in the breeding programme, preserving and "recycling" valuable genetic information even after these animals have left the food production system. This accelerates genetic improvement from young females and maximises the use of genetic information from animals destined for the food chain (Viana et al., 2018).

However, IVP has its own disadvantages and risks. IVP embryos show a higher incidence of pregnancy loss than IVV embryos (Holm & Callesen, 1998), with average pregnancy rates after transfer of about 40% for IVP compared with 64% for IVV embryos, a difference of about 24 percentage points (Ealy et al., 2019; Behrens et al., 2025), suggesting that the IVP process introduces factors that affect embryo viability and development. Embryonic loss may occur because of problems during conceptus elongation (Clemente et al., 2011; Hansen, 2020). Stressors in the in vitro environment, such as chemical exposure, mechanical handling, culture media, oxygen concentration and temperature fluctuations, can alter DNA methylation and induce oxidative stress, potentially leading to lasting changes in the embryo's transcriptome (de Barros & Paula-Lopes, 2018). The main molecular differences between IVP and IVV embryos arise during the in vitro period, which includes embryonic genome activation (EGA), a critical event for embryo viability (Khan et al., 2012). Culture conditions during IVP affect EGA and, in turn, the embryonic epigenome, transcriptome and proteome (Banliat et al., 2022; Gad et al., 2012; Salilew-Wondim et al., 2018).

Successful pregnancy requires a well-regulated dialogue between the developing embryo and the maternal environment (Chankeaw et al., 2021). After natural mating or artificial insemination (AI), the embryo reaches the uterus around day 4. In embryo transfer, embryos are placed in the uterus of a synchronised recipient on day 7. IVV embryos are traditionally produced by stimulating a donor cow's ovaries, leading to multiple ovulations, followed by AI, embryo collection and transfer. In IVP, oocytes are collected from live cows by transvaginal follicle aspiration or from ovaries after slaughter, and are matured, fertilised and cultured in vitro (Holm & Callesen, 1998). By day 7, most embryos have reached the blastocyst stage, which consists of a fluid-filled cavity and two cell types: the inner cell mass and the outer trophectoderm (TE). The inner cell mass forms the epiblast, which becomes the embryonic disc (ED) and eventually the fetus, and the hypoblast, which together with the TE forms the extra-embryonic membranes (EEM) that give rise to the placenta. Around days 9–10, the blastocyst hatches from the zona pellucida. The TE then proliferates rapidly and the conceptus changes from spherical to ovoid, tubular and finally filamentous by days 12–16, a process called conceptus elongation. Elongation increases the surface area of the TE, which is needed to secrete sufficient interferon tau (IFNT). IFNT prevents luteolysis, the regression of the corpus luteum, and thereby maintains the progesterone (P4) production needed for early pregnancy. In cattle, implantation occurs only after elongation. Higher P4 early in pregnancy accelerates conceptus elongation and increases pregnancy success, whereas low P4 delays the changes in the endometrial transcriptome needed to support the conceptus (Clemente et al., 2009).

Transcriptomic studies show that IVP and IVV blastocysts have distinct transcriptomic profiles (Rabaglino et al., 2021), and that embryos able to establish pregnancy differ in their transcriptome from those that are lost (Zolini et al., 2020); on the maternal side, the endometrial transcriptome also differs with uterine receptivity (Rabaglino & Kadarmideen, 2020). Conceptus length around day 15 is also associated with the transcriptome and with viability (Berg et al., 2010; Wang et al., 2009): short conceptuses exposed to low P4 fail to up-regulate many IFNT-dependent

and IFNT-independent endometrial genes, and longer conceptuses are more likely to establish pregnancy. IVP conceptuses are smaller than IVV conceptuses at day 13 (Clemente et al., 2011), indicating impaired or delayed initiation of elongation. In the full cohort from which the present samples were taken, IVP conceptuses were shorter than IVV conceptuses at day 15, and an embryonic disc was found in only 55% of IVP compared with 94% of IVV conceptuses (Behrens et al., 2025). Restricting the comparison to conceptuses of the same length reduces, but does not remove, differences in developmental stage. It also selects a particular subset: conceptuses that elongated to a given length and had a detectable disc, so that results apply to these selected conceptuses rather than to IVP conceptuses in general.

The aim of this project was to characterise ED and EEM expression differences between selected IVP and IVV day-15 conceptuses of the same length, to assess whether the group contrast differs between tissues, and to examine how sensitive the biological interpretation is to cohort selection, to individual embryos and to an expression-derived disc-enrichment score. Specifically, the objectives were to: (1) identify genes differentially expressed between the IVP and IVV groups in each tissue; (2) test whether the IVP–IVV contrast differs between ED and EEM, using a model that accounts for both tissues coming from the same embryo; (3) characterise the gene-set patterns involved; and (4) assess which patterns remain consistent across within-cohort sensitivity analyses.

Results

Samples, design constraints and global structure

The primary comparison comprised ED and EEM from four IVP conceptuses (IVP9, IVP10, IVP11, IVP14) and four IVV conceptuses (VE24, VE26, VE28, VE29) of the same length, recorded in the sample sheet as “elongated, short”; one additional IVP disc without a matching EEM library (IVPF10_EC) was excluded. Every library of the cohort, with its inclusion in each analysis, is listed in Supplementary Table S1. Measured conceptus lengths and embryo-to-cow assignments were not available for this analysis, so the matching tolerance and the cow of origin of each conceptus cannot be reported. Library sizes were uniform (55.6–57.4 million assigned reads). In PCA, the first component separated ED from EEM and the second separated IVP from IVV in both tissues (Figure 1). EEM library IVP9_T1A correlated less with the other libraries of its group than any other EEM library (mean r = 0.84 vs 0.91–0.97); its influence is examined under “Influence of individual embryos”.

Embryo sex was inferred from the strongly sex-dimorphic expression of XIST, which is much higher in female than in male conceptuses. All IVP libraries of the primary comparison had XIST below 1 count per million (inferred male), whereas all IVV libraries had 29–57 counts per million (inferred female), with ED and EEM of each conceptus in agreement (Supplementary Table S1). Y-linked genes could not serve as a second check: the only Y-annotated gene in the count matrix, RPS4Y1, was expressed at similar levels in all libraries. Production system and inferred sex are therefore fully confounded in the primary comparison.

To see which recorded sample characteristics drive the main axes of variation, the scores of the first ten principal components were correlated with the metadata of all 32 libraries in the cohort, and separately for the ED and EEM libraries (Supplementary Figure S3). Across all libraries, PC1 (61% of variance) was explained almost entirely by tissue (r = 0.95), so the difference between disc and membranes is by far the largest source of variation, which supports analysing the two tissues separately. Production group was the next-largest source of variation, associated with PC2 (12%; r = 0.62); this component also tracked library size (r = −0.67), indicating that part of the variation along it is technical. PC3 (4%) separated the same-length libraries from the rest of the cohort (r = 0.78), and embryo sex was associated with PC4 (3%; r = −0.82), on the same axis as production group because most IVP embryos were male.

Within the ED, PC1 (47%) was associated with production group (r = −0.76) and, along the same axis, with the disc-specific transcript score (r = 0.74): IVP disc sections scored low on both. Because the score is computed from the same expression data, this agreement is expected and is not independent evidence about tissue composition. Library size (r = 0.66) and sex (r = −0.53) also correlated with this component because they partly coincide with production group in these samples. Within the EEM, PC1 (35%) was associated with library size (r = −0.67) and production group (r = 0.58), and production group was also associated with PC3 (r = 0.58).

All IVP embryos were transferred fresh (Behrens et al., 2025). A sample-sheet label (“Vitrified”) nevertheless marks the four selected IVP conceptuses (IVP9, IVP10, IVP11 and IVP14) and no other conceptus; its meaning is unresolved. This label was associated with PC2 of the ED (r = 0.66) and, most strongly of any variable in the EEM, with PC3 of the EEM (r = 0.82), so the four embryos carrying it have a distinct expression profile, particularly in the membranes. Because the label is fully confounded with production group in the primary comparison, variation associated with the unresolved sample-sheet label cannot be separated from the IVP–IVV contrast. Overall, inferred sex, library size and the unresolved sample-sheet label all align with axes related to production group. These correlations are descriptive and do not establish cause, but they show that these factors cannot be separated from the IVP–IVV contrast in the primary comparison.

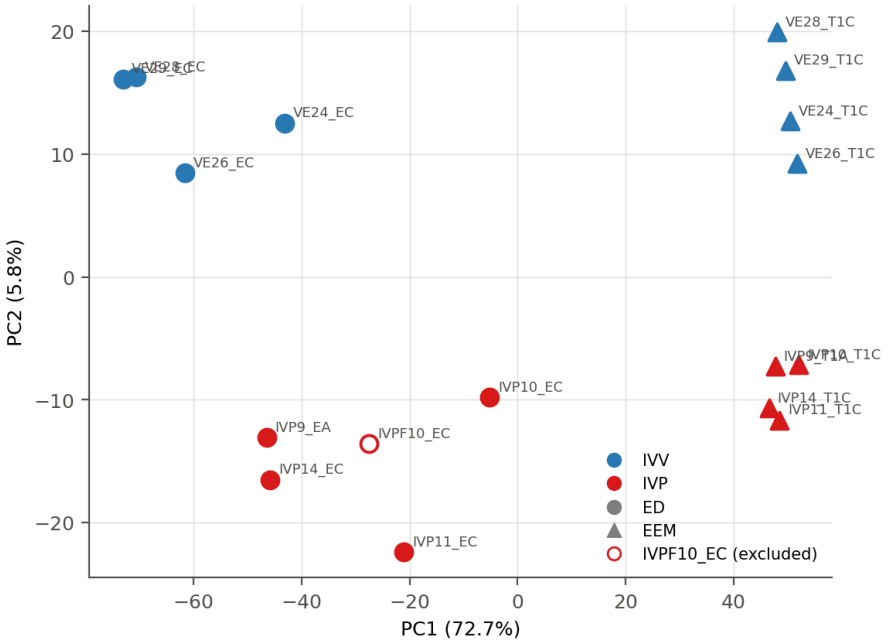


Figure 1. Principal component analysis of the 17 same-length libraries (log2 DESeq2-normalised counts, 500 most variable genes). Circles, embryonic disc (ED); triangles, extra-embryonic membranes (EEM); blue, IVV; red, IVP. The open circle is IVP disc IVPF10_EC (no matching EEM library), excluded from the primary analysis. PC1 separates the two tissues and PC2 separates IVP from IVV.

Differential expression between IVP and IVV groups in each tissue

Differential expression was tested on raw read counts with DESeq2 separately for each tissue (4 IVP vs 4 IVV embryos). In the ED, 1,218 of 15,641 tested genes were differentially expressed (FDR < 0.05): 467 higher and 751 lower in IVP. In the EEM, 721 of 14,564 tested genes were differentially expressed: 423 higher and 298 lower in IVP (Figure 2). Because all IVP embryos were male and all IVV embryos female, these DEGs are differences between the two groups, not effects of the in vitro procedure; X-linked genes, for example, were over-represented among them (see “Contribution of X-linked genes”).

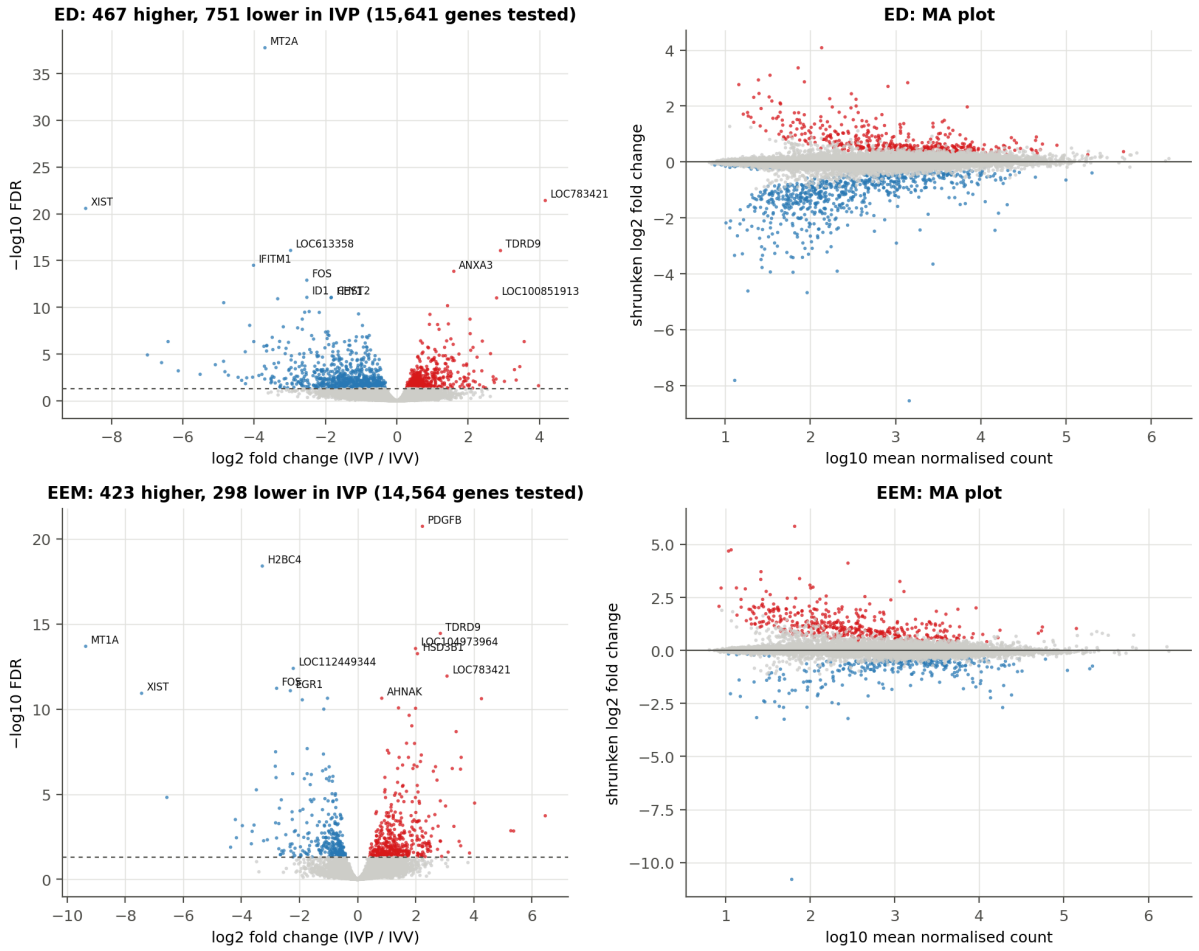


Figure 2. Differential expression between IVP and IVV embryos in the embryonic disc (top) and extra-embryonic membranes (bottom). Left: volcano plots of all tested genes; red, higher in IVP; blue, lower in IVP (Benjamini–Hochberg FDR < 0.05); dashed line, FDR = 0.05; the twelve most significant genes are labelled. Right: MA plots of apeglm-shrunken log2 fold changes.

The IVP–IVV contrast in the two tissues

ED had more DEGs than EEM (1,218 vs 721), but the median absolute shrunken log2 fold change across the 14,401 genes tested in both tissues was numerically similar (0.078 in ED and 0.079 in EEM); this descriptive summary does not establish that the two tissues are equally susceptible. The IVP–IVV contrasts of the two tissues were positively but moderately correlated (Pearson r = 0.46), and 295 genes were differentially expressed in both (Figure 3). To test formally whether the group contrast differs between tissues, a tissue × group interaction was fitted in a model that accounts for ED and EEM coming from the same embryo (but not for embryos sharing a cow). For 417 genes, the IVP–IVV contrast differed significantly between tissues (FDR < 0.05), so a subset of genes shows a tissue-dependent group contrast. This interaction model was not adjusted for the disc-specific transcript score, and part of the interaction reflects the lower disc-specific signal in IVP ED sections described below.

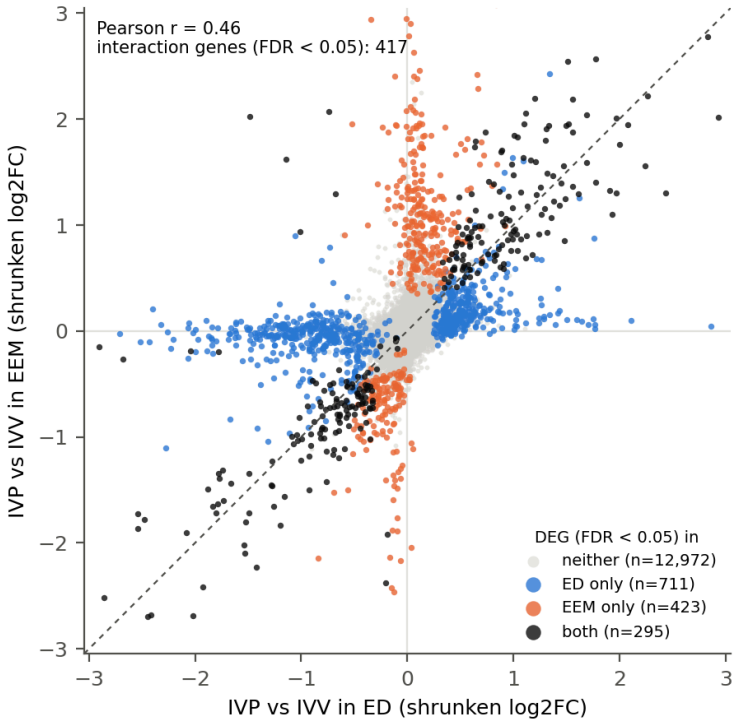


Figure 3. IVP vs IVV log2 fold change (apeglm-shrunken) for each gene in the ED (x) and EEM (y). Colour shows the tissue(s) in which the gene is a DEG (FDR < 0.05). Dashed line, equal effect in both tissues.

Gene-set patterns

Two complementary approaches were used. Over-representation analysis (ORA) tested whether GO biological-process terms occur among the up- or down-regulated DEGs more often than expected relative to all genes tested. Gene set enrichment analysis (GSEA) used every tested gene ranked by the DESeq2 Wald statistic and tested whether the members of a GO term or KEGG pathway cluster toward the top (higher in IVP) or bottom (lower in IVP) of the ranking. Because GSEA does not depend on a significance cut-off, it detected many more coordinated expression differences than ORA (Table 1). Enrichment describes coordinated expression of pathway-associated genes, not measured pathway activity. For the ED, results are shown both without and with adjustment for the disc-specific transcript score, an estimate of how much embryonic-disc tissue each ED section contains; how the two are read together is explained in "Developmental expression contrasts are sensitive to disc-score adjustment". The main terms are summarised in Figure 5, the strongest terms of each analysis are listed in Supplementary Table S2, and complete tables are provided with the analysis code.

Table 1. Number of significant GO biological-process terms and KEGG pathways (BH FDR < 0.05) per analysis. ORA, among DEGs higher (up) or lower (down) in IVP; GSEA, gene sets shifted toward higher or lower expression in IVP. For the interaction, "higher" means a more positive IVP–IVV contrast in the EEM than in the ED.

Analysis	GO ORA (DEGs)		GO GSEA		KEGG GSEA	
	up	down	higher	lower	higher	lower
ED	4	8	35	45	11	29
ED, disc-adjusted	12	0	116	44	19	24
EEM	0	6	123	95	39	35
Interaction (EEM − ED)	–	–	262	187	34	38

Embryonic disc gene sets

Among ED DEGs, genes higher in IVP were over-represented in glycosylation (15 genes, FDR = 0.009) and related glycoprotein-biosynthesis terms, and genes lower in IVP in anatomical structure morphogenesis (46 genes, FDR = 0.001), animal organ morphogenesis, neuron differentiation, enzyme-linked receptor signalling and protein refolding (FDR 0.005–0.043; Figure 4). GSEA identified 35 GO terms higher and 45 lower in IVP. Gene sets higher in IVP were dominated by protein modification and lipid metabolism: protein lipidation and palmitoylation (NES 2.16–2.29), glycosylation (2.15), and membrane-lipid and sphingolipid metabolism (1.97–2.00). The KEGG analysis agreed, with glycerophospholipid, glycerolipid, ether-lipid and sphingolipid metabolism, N-glycan and mucin-type O-glycan biosynthesis, and steroid hormone biosynthesis higher in IVP (NES 1.72–2.09). Gene sets lower in IVP were developmental: embryo development, animal organ morphogenesis, cell fate commitment and embryonic morphogenesis (NES −1.87 to −2.05) and canonical Wnt signalling (−1.66), together with chaperone-mediated protein folding (−1.88 to −1.98). In KEGG, TGF-β, Wnt and Notch signalling and signalling pathways regulating stem-cell pluripotency were lower in IVP (NES −1.74 to −1.80). Several KEGG disease pathways (for example, transcriptional misregulation in cancer and basal cell carcinoma) were also lower; they share developmental transcription factors and signalling genes with the pathways above and are not interpreted as disease processes.

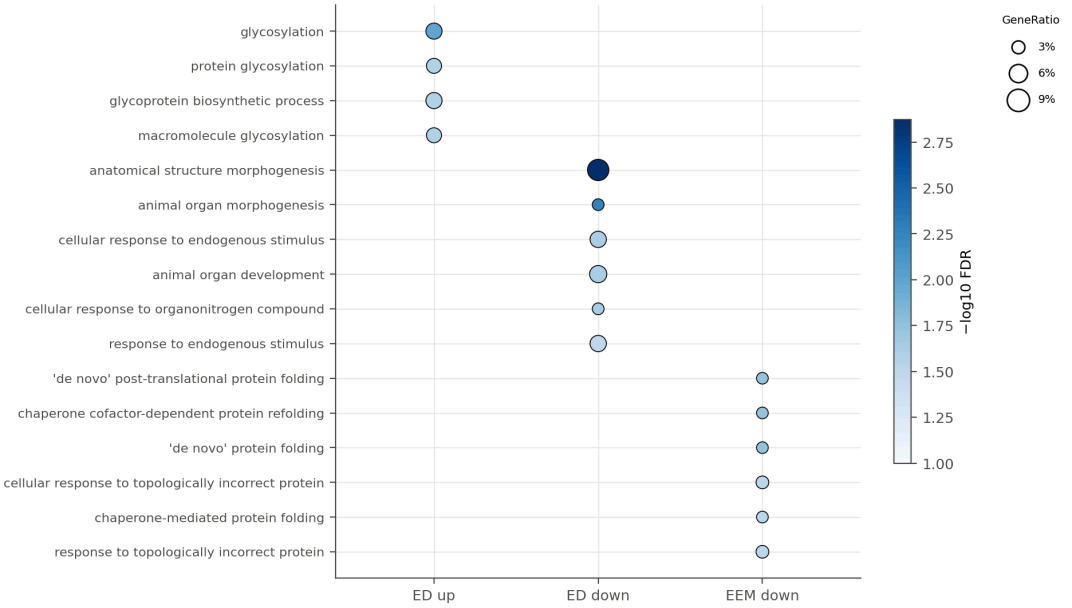


Figure 4. GO biological-process over-representation among genes higher (up) or lower (down) in IVP in each tissue. Background, all genes tested in that tissue; BH FDR < 0.05; the six most significant terms per set are shown. GeneRatio = DEGs annotated to the term divided by DEGs with any GO annotation. Genes higher in IVP EEM gave no significant terms.

Extra-embryonic membrane gene sets

Among EEM DEGs, ORA found no terms for genes higher in IVP and six protein-folding terms for genes lower in IVP (5–6 chaperone genes, FDR 0.027–0.043; Figure 4). GSEA was more informative, with 123 GO terms higher and 95 lower in IVP. Gene sets higher in IVP EEM concerned the cell surface and the extracellular environment: cell adhesion and cell migration (NES 1.94 and 1.85), wound healing and blood coagulation (2.10 and 1.98), axon guidance and semaphorin–plexin signalling (2.10 and 1.96), and potassium-ion transport (2.09). In KEGG, ECM–receptor interaction, integrin signalling, cell adhesion molecules, cytoskeleton in muscle cells, calcium signalling, cytokine–cytokine receptor interaction, complement and coagulation cascades, and glycosphingolipid biosynthesis were higher in IVP (NES 1.80–2.21). Gene sets lower in IVP EEM concerned energy metabolism, protein synthesis and protein handling: aerobic respiration, oxidative phosphorylation and the electron transport chain (NES −2.25 to −2.33), ATP biosynthesis (−2.14), cytoplasmic translation and ribosome biogenesis (−2.14 and −1.96), and protein folding and the response to unfolded protein (−2.14 to −2.21). In KEGG, protein processing in the endoplasmic reticulum (−2.48), ribosome (−2.35), proteasome, the citrate (TCA) cycle and spliceosome (−1.99 to −2.16) were lower. The ad-

hesion and extracellular-matrix pattern kept its direction in the within-cohort sensitivity analyses, whereas the translation and respiration pattern did not (see “Within-cohort sensitivity analyses”).

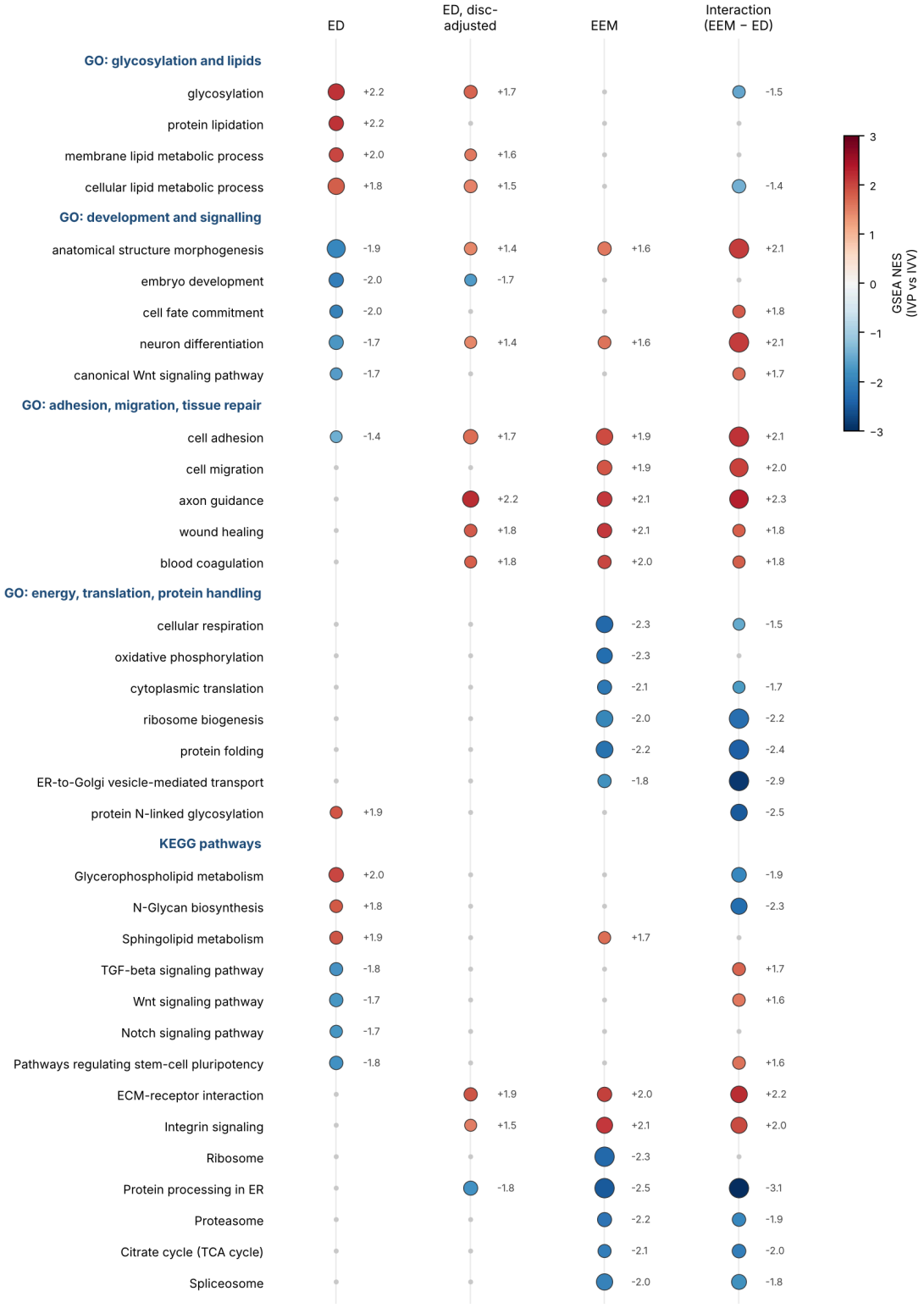


Figure 5. Overview of GSEA results for representative GO biological-process terms and KEGG pathways. Columns: IVP vs IVV in the ED, in the ED adjusted for the disc-specific transcript score, and in the EEM; and the tissue × group interaction (IVP–IVV contrast in EEM minus that in ED). Colour and number, normalised enrichment score (NES; red, higher in the IVP group or a more positive contrast in the EEM); circle size increases with −log10 FDR; small grey dot, not significant (FDR ≥ 0.05).

Gene sets whose group contrast differs between tissues

GSEA of the interaction statistic, which ranks genes by how much the IVP–IVV contrast in the EEM exceeds that in the ED, identified 262 GO terms with a more positive and 187 with a more negative contrast in the EEM. Terms with a more positive contrast in the EEM were adhesion and morphogenesis programmes: axon guidance and axon development (NES 2.31–2.34), cell adhesion (2.14), generation of neurons and cell morphogenesis (2.09–2.11), and in KEGG, ECM–receptor interaction, cell adhesion molecules, cytoskeleton in muscle cells, integrin signalling and axon guidance (1.96–2.22). Figure 5 shows that this contrast arises because these programmes were lower in IVP ED but higher in IVP EEM; after adjusting the ED for the disc-specific transcript score, they were higher in IVP in both tissues. Terms with a more negative contrast in the EEM formed the secretory pathway: endoplasmic reticulum-to-Golgi vesicle transport (NES −2.86), Golgi vesicle transport (−2.63), protein N-linked glycosylation (−2.51), protein folding (−2.44), protein targeting to the endoplasmic reticulum and the response to endoplasmic-reticulum stress (−2.31 to −2.38), and in KEGG, protein processing in the endoplasmic reticulum (−3.06), N-glycan biosynthesis (−2.29) and protein export (−2.29). This contrast combines higher glycosylation and lipid-metabolism genes in IVP ED with lower protein-processing genes in IVP EEM.

Developmental expression contrasts are sensitive to disc-score adjustment

Because the disc is small relative to the surrounding membranes, every ED section also contains some EEM tissue. As an expression-based indicator of disc content, ED-enriched marker genes were defined from the four IVV conceptuses of the primary comparison (ED vs EEM of the same conceptus), and a disc-specific transcript score was computed as their mean VST expression. The score is not a measured cell proportion, and because its markers were selected in the comparison group itself, it is not independent of the IVP–IVV comparison. It is relevant because an ED section with less disc tissue would show lower expression of disc-enriched genes even if every disc cell expressed them normally, through dilution by membrane cells; lower expression of disc-identity genes within disc cells would produce the same pattern. IVP ED sections had lower scores than IVV ED sections (7.3–8.5 vs 8.3–9.3); IVP10_EC and IVP11_EC were lower than every IVV section (Figure 6A). Across ED DEGs, the IVP fold change was strongly anti-correlated with disc specificity (Spearman $\rho = -0.66$), and 55.5% of the genes lower in IVP ED were disc-enriched genes.

After adjusting the ED model for the disc-specific transcript score, the number of genes lower in IVP fell from 751 to 299 (659 DEGs in total). The developmental terms that were lower in IVP ED became enriched among genes higher in IVP (NES +1.45 to +1.67), the same direction as in the EEM, whereas glycosylation and lipid metabolism remained higher in IVP (Figure 6B). Not every developmental signal disappeared: embryo development remained lower in IVP (NES −1.68), and gene sets for cell cycle, ubiquitin-dependent protein catabolism, mRNA processing and the DNA damage response were lower in IVP only after adjustment (NES −1.49 to −1.88; KEGG p53 signalling, apoptosis and cell cycle). At the gene level, most ED Wnt, BMP and TGF-β DEGs (e.g. WNT3, WNT5A, LEF1, BMP4, DKK3, TGFB2) tracked the disc-specific transcript score and lost significance after adjustment (Figure 7). This sensitivity is compatible with less disc tissue in IVP sections, but also with differences in disc-cell state or in marker expression, which bulk RNA-seq cannot distinguish.

How the two ED analyses are read. The unadjusted model keeps all group differences but cannot separate changes in expression from differences in how much disc material each section contains. The score-adjusted model is an exploratory sensitivity analysis conditional on an expression-derived covariate: because the score is itself lower in the IVP group and is built from disc-enriched genes, adjustment may remove both composition-associated and biologically regulated differences, and its coefficients are not effects at experimentally verified equal tissue composition. The two models are therefore read together. A gene set that differs in both (glycosylation, lipid metabolism) is a difference retained after adjustment for the disc-enrichment score; a gene set that differs only without adjustment (most morphogenesis, Wnt and BMP terms) is sensitive to the score; and a gene set that differs only after adjustment (cell cycle, DNA damage response) is exploratory. Neither pattern establishes or excludes a compositional explanation. Lower disc-enriched expression in IVP sections is itself a group difference and is reported, not discarded.

Dependence on marker selection. Rebuilding the disc-marker set four times, each time without one of the four IVV conceptuses, gave 262–471 markers that included nearly all original markers (261–263 of 263) and scores almost identical to the original ($r \geq 0.999$). In every version, the IVP group had a lower mean score and two IVP sections scored below every IVV section, the score-adjusted model found 607–663 DEGs (659 originally), and the adjusted gene-set results were unchanged (glycosylation NES 1.73–1.76, anatomical structure morphogenesis 1.37–1.46, embryo development −1.71 to −1.80). The score is therefore insensitive to which IVV conceptus defines it, although this does not show that it measures tissue composition; its dependence on individual IVP conceptuses is described under “Influence of individual embryos”.

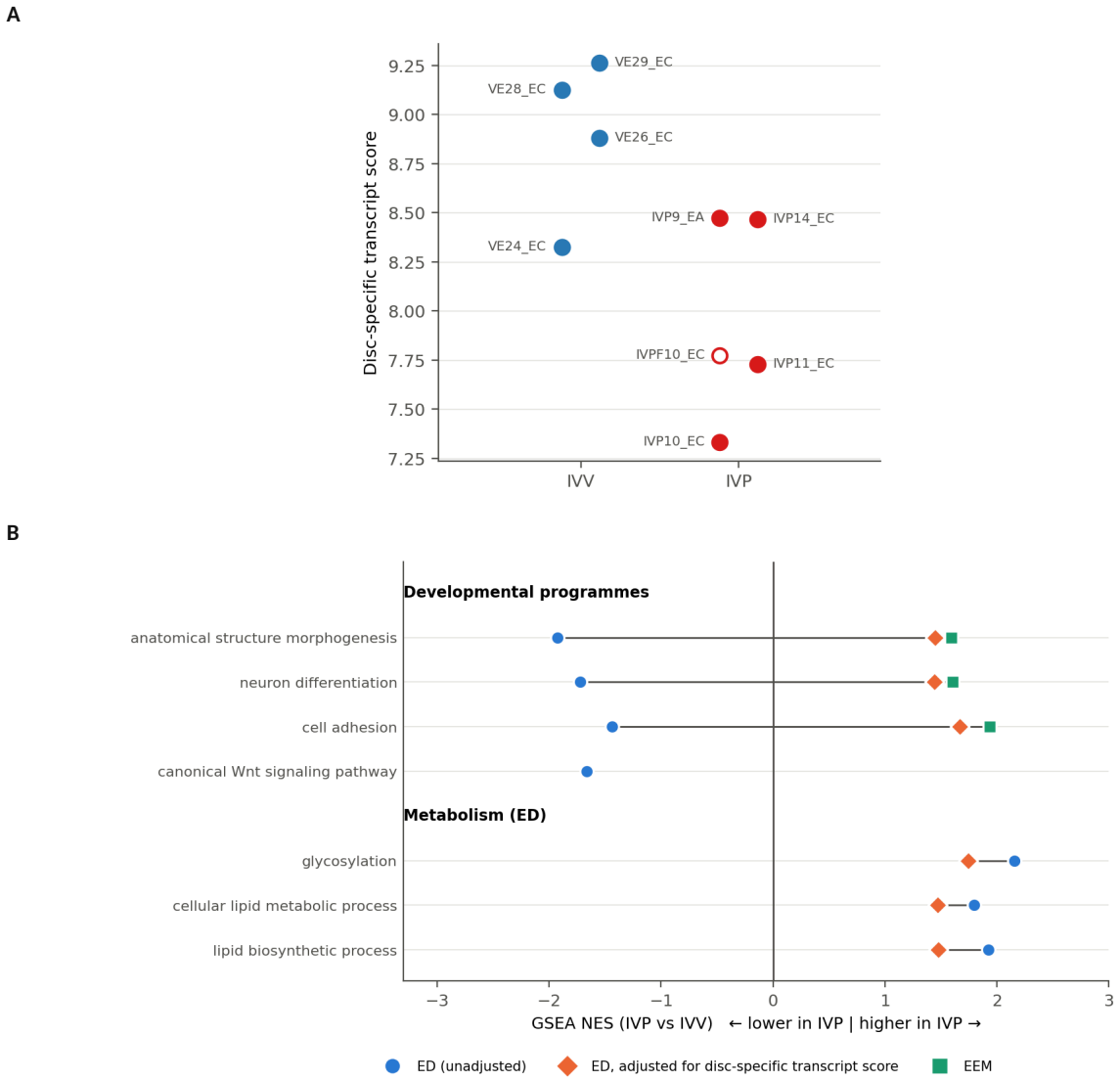


Figure 6. (A) Disc-specific transcript score (mean VST expression of ED-enriched marker genes, defined from IVV embryos only) for each ED section; open circle, excluded IVPF10_EC. (B) GSEA normalised enrichment scores (IVP vs IVV) for selected GO terms in the ED without and with adjustment for the disc-specific transcript score, and in the EEM. Only terms significant at FDR < 0.05 are drawn; a missing marker means not significant in that analysis.

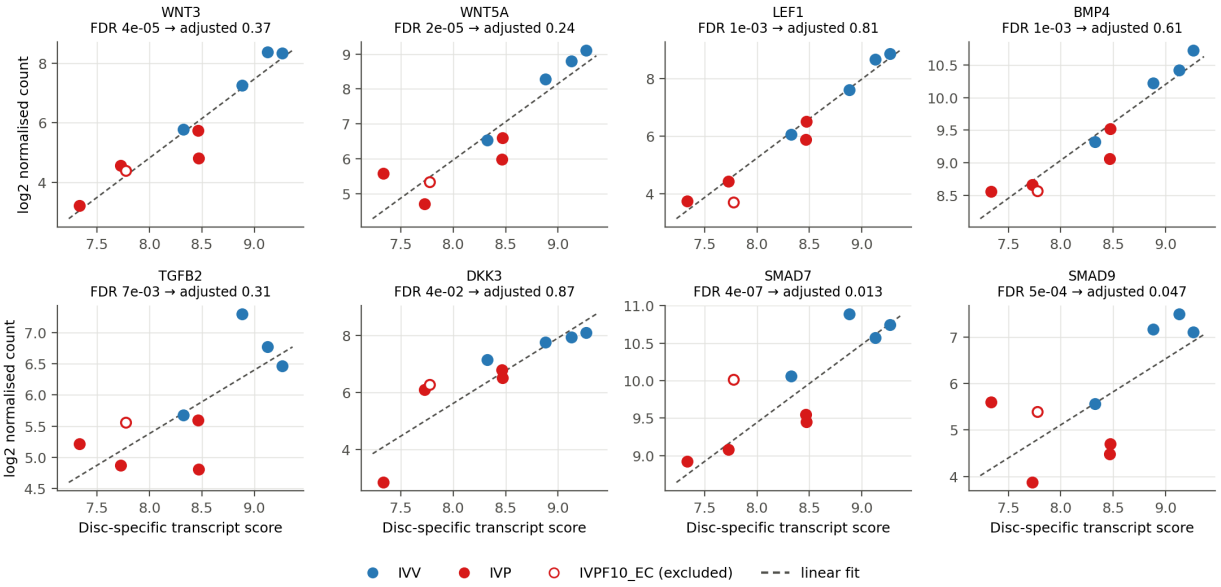


Figure 7. Expression of ED Wnt, BMP and TGF-β DEGs against the disc-specific transcript score. Dashed line, linear fit across the eight libraries. Titles give the FDR for IVP vs IVV before and after adjusting for the disc-specific transcript score.

Wnt, BMP/TGF-β and Notch signalling genes

To show the developmental signalling changes at the level of individual genes, ED DEGs from the leading edge of the KEGG Wnt, TGF-β and Notch pathways were plotted for both tissues (Figure 8). In the ED, all 17 genes were lower in IVP (log2 fold change −1.1 to −3.7). Ten of them lost significance after adjustment for the disc-specific transcript score (adjusted FDR 0.09–0.81): WNT3, WNT5A, LEF1, NKD1, NOTCH1, DLL3 and MFNG, which in IVV embryos are expressed much more highly in the disc than in the membranes, and BMP4, ID4 and TGFB2. Their contrast is therefore sensitive to the disc score. Six genes remained lower in IVP after adjustment: WNT10B, FRAT2, ID1, SMAD7, SMAD9 and HEY1 (adjusted FDR 0.0001–0.047), and DTX1 was close (0.098).

Two patterns stood out among these genes. First, the Notch target HEY1, the Wnt-pathway gene FRAT2 and the Wnt feedback inhibitor NKD1 were also lower in IVP EEM (FDR ≤ 3×10^{-5}), so their lower expression in the IVP group is shared by both tissues rather than specific to the disc. HEY1 was lower in the IVP group in both tissues in the male-only and full-cohort comparisons as well, and the BMP target ID1 was lower in IVP EEM in both of these analyses (FDR 0.044 and 0.001), making HEY1 and ID1 the most consistent signalling differences. Second, the disc-specific reductions that survived adjustment in the primary comparison (WNT10B, SMAD7, SMAD9) did not survive adjustment in the full cohort (FDR 0.45–0.84), although SMAD9 was lower in IVP ED in the unadjusted male-only and full-cohort comparisons. WNT5A showed opposite trends in the two tissues (lower in IVP ED, higher in IVP EEM; EEM FDR 0.10). The per-gene results for all analyses are listed in the repository table signalling_genes.csv.

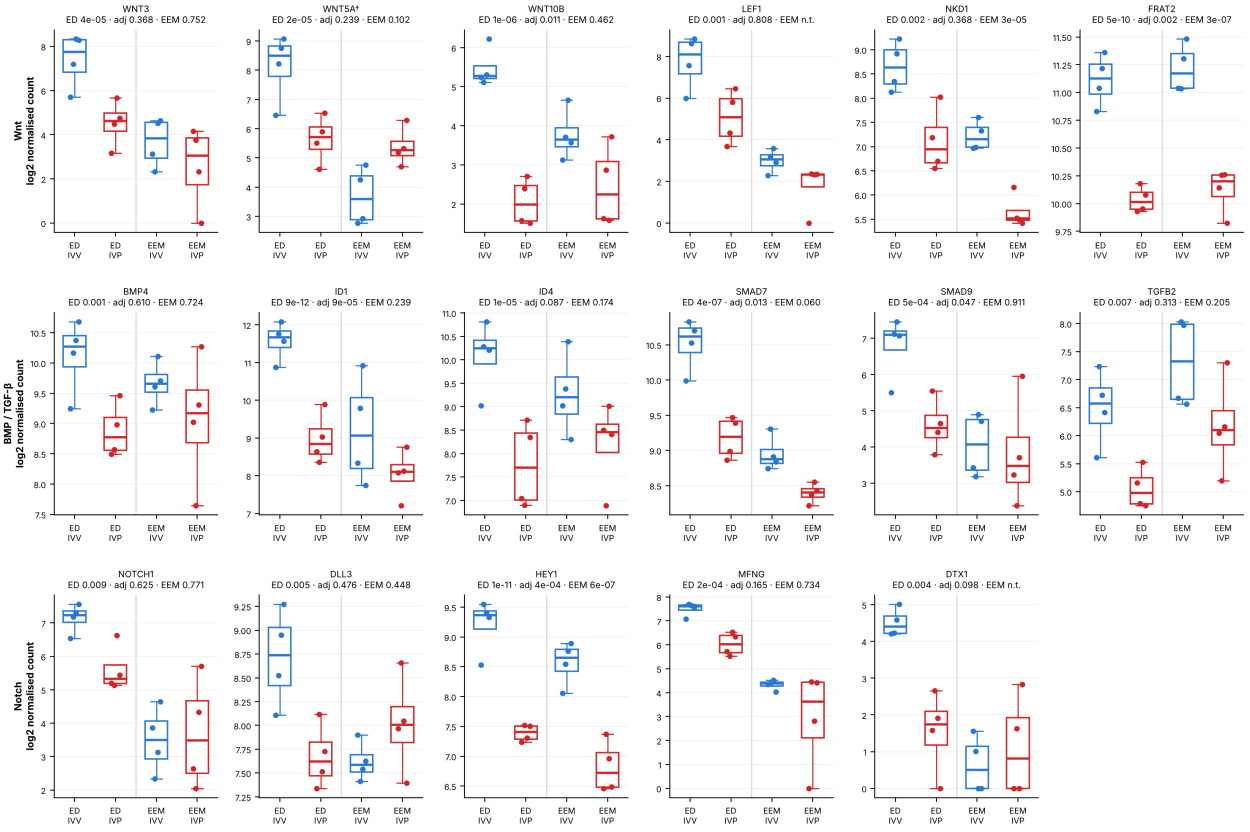


Figure 8. Expression of Wnt (top), BMP/TGF-β (middle) and Notch (bottom) genes in the ED and EEM of IVV (blue) and IVP (red) embryos; one point per embryo, log2 DESeq2-normalised counts. Genes are ED DEGs from the leading edge of the corresponding KEGG pathway. Titles give the FDR for IVP vs IVV in the ED, in the ED adjusted for the disc-specific transcript score (adj) and in the EEM; n.t., not tested (too few reads in the EEM). †, disc marker gene used to compute the disc-specific transcript score.

Genes driving the main gene-set patterns

Many of the significant GO terms and KEGG pathways overlap, so several labels may describe the same few genes. For each main pattern, the leading-edge genes of a representative gene set and of related sets were therefore compared (Table 2). The glycosylation sets overlapped strongly (median Jaccard 0.61): glycosylation, protein glycosylation, glycoprotein biosynthesis and N-glycan biosynthesis describe one signal, carried by genes such as SLC35C1, A4GALT, FUT2, ALG11, MOGS and MGAT1. The lipid-metabolism sets overlapped less (0.16) and were driven by sphingolipid and fatty-acid genes (SPTSSA, CERS5, ELOVL6, DEGS2, B4GALNT1), so lipid metabolism is a broader pattern than glycosylation. The developmental sets overlapped little (0.03) apart from shared Wnt ligands (WNT3, WNT5A, WNT5B, WNT10B, WNT16, WNT3A), and 15% of the morphogenesis leading edge were disc-score marker genes, compared with none in the glycosylation and lipid leading edges. In the EEM, adhesion sets shared integrin and matrix genes (ITGB3, ITGB6, ITGB7, ITGA2, LAMA1, THBS4, VWF), whereas translation and respiration sets did not overlap and represent separate gene groups.

Most leading-edge genes kept the sign of their fold change in the disc-score-adjusted, male-only and full-cohort analyses (66–88%). The exceptions were the EEM translation/respiration leading edge (39–43%, i.e. mostly reversed) and protein folding (53–56%), whose leading edge includes many chaperones besides the consistently lower inducible heat-shock genes. For the developmental leading edge, 78% of genes kept a negative fold change after disc-score adjustment even though the gene set as a whole moved to a positive NES. Because GSEA ranks each gene against all others, a set can change sign when adjustment shifts the whole ranking while its own genes are only weakened; the adjusted developmental result is therefore best described as a loss, not a reversal, of the group contrast.

Table 2. Leading-edge (core-enrichment) genes behind the main gene-set patterns. Themes, representative sets and related sets were fixed before the analysis. Representative set: GO term and number of leading-edge genes. Related sets: number of GO terms/KEGG pathways in the theme and the median pairwise Jaccard overlap of their leading edges (1 = identical genes, 0 = no shared genes). Recurrent drivers: leading-edge genes shared by at least half of the related sets, ordered by the size of the group contrast. Disc markers: share of the representative leading edge that belongs to the disc-score marker set. Same direction: share of representative leading-edge genes whose fold change has the same sign as in the primary analysis in the disc-score-adjusted ED / male-only / full-cohort analyses. For protein folding in the ED the representative set is protein refolding.

Theme	Representative set (genes)	Related sets (overlap)	Recurrent drivers	Disc markers	Same direction: adjusted / male-only / full
ED glycosylation	glycosylation (54)	4 (0.61)	SLC35C1, A4GALT, FUT2, ALG11, DOLPP1, MOGS, MGAT1, CRPPA	0%	85% / 70% / 81%
ED lipid metabolism	cellular lipid metabolic process (150)	5 (0.16)	SPTSSA, TRAM2, CERS5, PIGO, B4GALNT1, ST6GALNAC6, ELOVL6, DEGS2	0%	85% / 67% / 67%
ED development / signalling	anatomical structure morphogenesis (144)	6 (0.03)	WNT10B, WNT5A, WNT3, TBX6, WNT5B, BAMBI, WNT16, WNT3A	15%	78% / 87% / 88%
EEM adhesion / ECM	cell adhesion (92)	5 (0.12)	ITGB7, ITGA2, TYRO3, ITGB3, LAMA1, THBS4, ITGB6, VWF	–	– / 76% / 66%
EEM translation / respiration	cytoplasmic translation (28)	5 (0.00)	RPSA, RPLP0, LOC781576, RPL26	–	– / 43% / 39%
Protein folding (ED and EEM) – ED	protein refolding (9)	2 (0.29)	HSPA1A, HSPA6, HSPA8, HSPA2	0%	100% / 56% / 56%
Protein folding (ED and EEM) – EEM	protein folding (64)	4 (0.27)	DNAJB1, HSPA6, HSPA8, DNAJB5, HSPA1A, DNAJB2, DNAJB13, DNAJA4	–	– / 53% / 55%

Heat-shock genes

The inducible heat-shock chaperones that drive the protein-folding terms (HSPA1A, HSPA6, DNAJB1) and HERPUD1 were lower in the IVP group in both tissues (Figure 9), and their direction was the same in all sensitivity analyses (for example, HSPA6 log2 fold change −1.6 to −5.6). These genes also respond to handling stress, so this shared pattern may partly reflect differences in collection; its cause is unresolved.

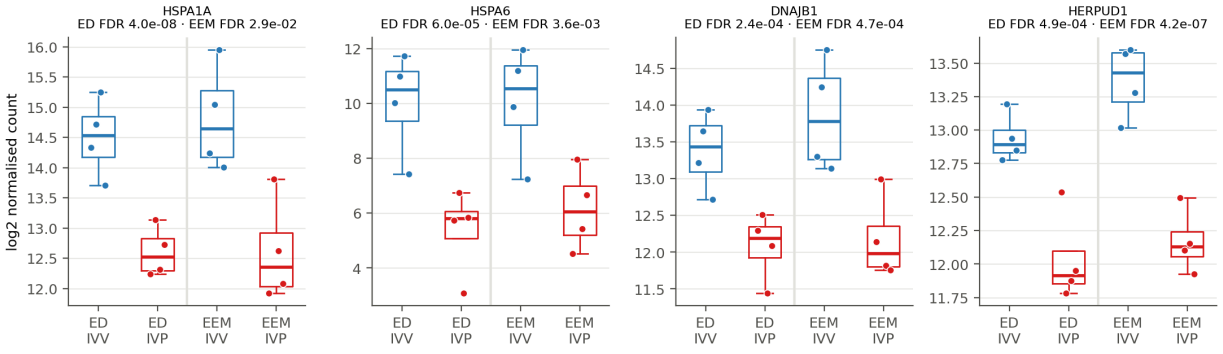


Figure 9. Expression of heat-shock and protein-folding genes in ED and EEM of IVV and IVP embryos; one point per embryo. Titles give the FDR for IVP vs IVV in each tissue.

Within-cohort sensitivity analyses

The analyses in this and the following sections reuse samples from the primary comparison and the same study conditions. They test how much the interpretation depends on particular samples, models or embryos and are not independent replications. Two of them used additional embryos from the same cohort (Materials and methods). In a comparison of male embryos only (ED, 7 IVP vs 2 IVV; EEM, 5 vs 2; the 2 IVV males are not part of the primary comparison), 78% of the ED DEGs and 80% of the EEM DEGs from the primary analysis changed in the same direction. In a full-cohort model including embryos of all lengths with sex and length group as covariates (16 ED and 14 EEM libraries, including all primary libraries), 76% and 78% changed in the same direction. Across all tested genes, fold changes correlated moderately with the primary analysis (Spearman ρ = 0.36–0.42).

At the gene-set level (Figure 10), glycosylation and lipid metabolism were higher in IVP ED in every analysis and significant in the full cohort with and without adjustment for the disc-specific transcript score (NES 1.4–1.9); in the male-only comparison, which has only two IVV embryos, they had the same direction but were not significant. The developmental terms were lower in IVP ED in the male-only and full-cohort analyses (NES −2.2 to −2.4) and, as in the primary analysis, were no longer lower after disc adjustment. In the EEM, translation, ribosome biogenesis and cellular respiration reversed direction in both analyses (NES +1.5 to +2.6), so this pattern is not stable across cohort specifications; the source of the discrepancy (sex, cow, sample selection or model) is unresolved. Cell adhesion, cell migration and protein folding kept their direction but were not significant in the smaller or more heterogeneous analyses.



Figure 10. GSEA normalised enrichment scores for key GO terms in the primary (same-length) analysis and in two within-cohort sensitivity analyses: male embryos only, and the full cohort with sex and length group as covariates; for the ED, also with adjustment for the disc-specific transcript score. Black outline, FDR < 0.05; faded, not significant.

Influence of individual embryos

To test whether single conceptuses drive the main patterns, each of the eight conceptuses was omitted in turn (both of its libraries) and all four models were refitted (Figure 11). Across omissions, 99.8–100% of primary DEGs kept their direction, and fold changes correlated with the full analysis at Pearson r = 0.96–0.99 in the ED, 0.88–0.98 in the EEM and 0.92–0.97 for the interaction. In the unadjusted ED, glycosylation and lipid-metabolism gene sets stayed higher in the IVP group (NES 1.66–2.27) and the developmental gene sets stayed lower (NES −1.39 to −2.07) whichever conceptus was omitted. In the EEM, adhesion and migration (NES 1.73–2.02) and translation, ribosome

biogenesis, respiration and protein folding (NES −1.49 to −2.60) kept their direction in every omission, and the interaction patterns were equally stable. The inducible heat-shock genes and the signalling genes HEY1, ID1 and FRAT2 kept a negative fold change in every omission in both tissues.

The disc-score-adjusted ED model was less stable. When IVP14 was omitted, glycosylation and lipid-metabolism gene sets reversed (NES −0.79 to −0.92), translation-related gene sets changed sign, and the fold-change correlation with the full adjusted analysis fell to 0.85 (0.84 when IVP9 was omitted). IVP14 and IVP9 have the highest disc scores of the IVP group (both 8.47); a likely reason is that, without one of them, the score is more closely aligned with group membership, so that the adjusted group contrast is poorly determined. The adjusted ED result therefore depends on individual conceptuses, whereas the unadjusted ED and EEM patterns do not.

Omitting only the EEM library IVP9_T1A (equivalent to omitting IVP9 for the EEM model) increased the number of EEM DEGs from 721 to 805. All 709 comparable primary DEGs kept their direction, 547 remained significant, fold changes correlated at r = 0.88, and every EEM gene-set pattern kept its direction, so the EEM results are stable to this library.

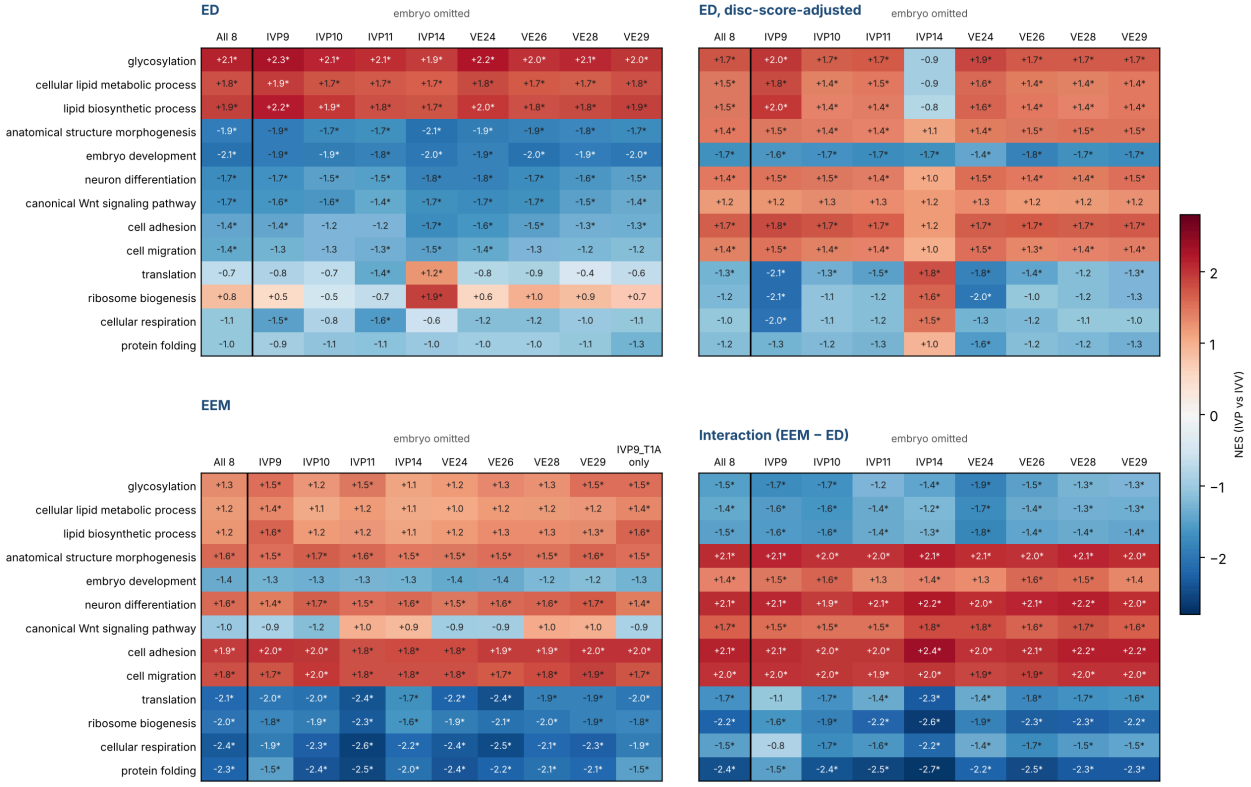


Figure 11. Influence of individual conceptuses on the main gene-set patterns. Each column gives the GSEA normalised enrichment score (NES) with all eight conceptuses (left of the vertical line) or with one conceptus omitted; * FDR < 0.05 among the 13 gene sets, which were fixed in advance. Red, higher in the IVP group (for the interaction, a more positive contrast in the EEM). NES were computed with fgsea on the same gene sets in every column and can differ slightly from the main GSEA. In the EEM panel, omitting IVP9 and omitting only library IVP9_T1A are the same model.

Contribution of X-linked genes

Because all IVP embryos in the primary comparison were male and all IVV embryos female, X-linked genes could appear to respond to IVP because of sex alone, for example through dosage differences arising from incomplete X inactivation. X-linked genes made up 3.4% of the genes tested in the ED and 3.2% in the EEM, but 8.0% of ED DEGs and 10.4% of EEM DEGs (Table 3; Fisher's exact test, odds ratio 2.8 and 4.0, $p < 10^{-15}$). Most X-linked DEGs were lower in IVP (79 of 97 in the ED and 55 of 75 in the EEM), as expected if they are more highly expressed in female embryos. They include XIST itself and genes such as KDM6A, USP9X and ZFX that escape X inactivation in other

mammals. These X-linked DEGs are therefore best regarded as sex-related rather than IVP-related; they are listed in the repository table xlinked_DEGs.csv.

Removing all X-linked genes and repeating GSEA changed the key pathway results very little. For the key GO terms of the ED, the disc-adjusted ED and the EEM (those shown in Figure 10, together with lipid biosynthesis and canonical Wnt signalling in the ED), the NES changed by at most 0.12 and every term stayed significant except neuron differentiation in the disc-adjusted ED, which moved just above the threshold (FDR 0.034 to 0.052). The pathway conclusions therefore do not depend on X-linked genes, although this check cannot exclude effects of sex on autosomal genes.

Table 3. X-linked genes among tested genes and DEGs (FDR < 0.05). Odds ratio and p-value, Fisher's exact test for over-representation of X-linked genes among DEGs. Chromosome location was unavailable for about 3% of genes.

Analysis	Genes tested	X-linked tested	DEGs	X-linked DEGs	X-linked DEGs higher / lower in IVP	Odds ratio	p
ED	15,641	528 (3.4%)	1,218	97 (8.0%)	18 / 79	2.8	9×10^{-16}
ED, disc-adjusted	15,660	529 (3.4%)	659	59 (9.0%)	19 / 40	3.0	7×10^{-12}
EEM	14,564	469 (3.2%)	721	75 (10.4%)	20 / 55	4.0	8×10^{-20}

Summary of sensitivity analyses

Table 4 brings the sensitivity analyses together. Because they reuse the same samples, agreement between them shows consistency within this dataset, not independent replication. Each pattern is assigned to one of three categories: consistent within this dataset (direction persists across the relevant analyses), analysis-sensitive (the interpretation changes with cohort, adjustment or omission of a conceptus) or unresolved (the data cannot distinguish competing explanations).

Table 4. Summary of the main patterns across analyses. Arrows give the direction in the IVP group (↑ higher, ↓ lower); numbers are GSEA NES (first and second term of the theme where two are given); n.s., FDR ≥ 0.05; n.a., not applicable. Full cohort: without / with the disc-specific transcript score. One conceptus omitted: number of the eight omissions in which the direction was kept.

Pattern	Primary	Disc-score-adjusted ED	Male-only	Full cohort	X-linked genes removed	One conceptus omitted	Assessment
ED glycosylation and lipid metabolism	↑ (2.15; 1.80)	↑ (1.74; 1.47)	↑ n.s. (1.21; 1.03)	↑ (1.90; 1.62) / ↑ (1.67; 1.44)	↑ (2.13; 1.79)	8/8 unadjusted; adjusted 7/8 (reversed without IVP14)	Consistent within this dataset; the adjusted version depends on IVP14
ED development (morphogenesis, Wnt)	↓ (−1.93; −1.66)	lost (+1.45; +1.20 n.s.); embryo development still ↓ (−1.68)	↓ (−2.30)	↓ (−2.43) / lost (−1.18 n.s.)	↓ (−1.96; −1.76)	8/8 unadjusted; adjusted: not lower in 8/8	Analysis-sensitive (depends on disc-score adjustment); cause unresolved
EEM adhesion and extracellular matrix	↑ (1.94)	n.a.	↑ n.s. (1.06)	↑ n.s. (1.23)	↑ (1.85)	8/8	Consistent in direction; weaker support outside the primary comparison
EEM translation and respiration	↓ (−2.11; −2.32)	n.a.	reversed ↑ (1.94; 1.79)	reversed ↑ (1.80; 1.53 n.s.)	↓ (−2.01; −2.22)	8/8	Analysis-sensitive (stable within the primary comparison, reverses with other embryos); source unresolved
Inducible heat-shock genes (HSPA1A, HSPA6, DNAJB1)	↓ in both tissues	↓	↓	↓ / ↓	n.a.	8/8	Consistent; cause (e.g. collection stress) unresolved
HEY1 and ID1	↓ in ED and EEM (ID1 EEM n.s.)	↓	↓ (HEY1 both tissues; ID1 EEM)	↓ / n.s.	n.a.	8/8	Consistent in direction
Lower disc-enriched expression in IVP ED sections	mean score 8.00 vs 8.90	n.a.	n.a.	n.a.	n.a.	lower in all 5 marker definitions	Consistent observation; meaning (amount of disc tissue vs disc-cell state) unresolved

Discussion

This analysis compared ED and EEM transcriptomes of selected IVP and IVV day-15 conceptuses of the same length. The two groups differed in the expression of 1,218 genes in the ED and 721 genes in the EEM, and for 417 genes the IVP–IVV contrast differed between tissues. Because the IVP group was entirely male, was carried by two recipients and carries an unresolved sample-sheet label, while the IVV group was entirely female and came from two superovulated donors, these are differences between two bundles of conditions rather than isolated effects of in vitro production. The findings are therefore discussed in three categories: patterns that were consistent across the within-cohort sensitivity analyses, patterns that depended on the analysis, and questions that the data cannot resolve.

Consistent patterns. The most consistent ED pattern was higher expression of glycosylation and lipid-metabolism genes in the IVP group. It kept its direction when any single conceptus was omitted, after removing X-linked genes, in the full cohort and, without reaching significance, in the male-only comparison. It was also retained after adjustment for the disc-enrichment score, but that adjusted result reversed when one IVP conceptus (IVP14) was omitted, so the claim that it is independent of tissue composition rests on a model that is itself unstable. The glycosylation signal is carried by one coherent group of glycosyltransferase and glycan-processing genes, whereas the lipid signal is broader. This agrees with earlier reports that carbohydrate- and lipid-metabolism terms are enriched among genes differing between IVP and IVV embryos (Rabaglino, 2023) and with altered lipid profiles of IVP blastocysts (Jana Idrissi et al., 2021); culture conditions such as glucose concentration and oxygen tension are known to affect IVP embryos (Berling et al., 2024). Whether such expression differences persist from culture to day 15, or reflect sex or maternal environment, cannot be decided here; metabolomic or lipidomic measurements of the disc in a balanced design would test them directly. In the EEM, adhesion, extracellular-matrix and integrin gene sets were higher in the IVP group and kept their direction across analyses, although with weaker support outside the primary comparison. Because the trophectoderm must attach to the uterine epithelium after elongation, these genes are candidates for follow-up, for example at the protein level. The Notch target HEY1 and the BMP target ID1 were the most consistent individual signalling genes, lower in the IVP group in both tissues.

Analysis-dependent patterns. Developmental gene sets (morphogenesis, Wnt, BMP/TGF-β and Notch) were lower in the unadjusted IVP ED comparison, but this contrast was largely lost after adjustment for the disc-enrichment score, both in the primary comparison and in the full cohort. Three explanations are compatible with this: IVP sections contained less disc tissue, disc cells expressed disc-identity genes at lower levels, or a mixture of developmental stage, regulation and sampling differences. The score cannot distinguish them, because it is an expression-derived quantity defined in the comparison group itself; loss of significance after adjustment does not prove dilution, and persistence after adjustment does not prove composition-independent regulation. A lower disc-enriched signal in IVP sections is nevertheless consistent with the impaired disc development in the full cohort, in which an embryonic disc was found in only 55% of IVP compared with 94% of IVV conceptuses (Behrens et al., 2025), and in vitro culture is known to alter cell-lineage composition already at the blastocyst stage (Ming et al., 2023). In the EEM, lower translation, ribosome-biogenesis and respiration gene sets reversed in the male-only and full-cohort analyses; this pattern is not stable, and its source is unresolved.

Relation to the parent study. Behrens et al. (2025) analysed the full cohort (7 IVP and 9 IVV embryonic discs, corrected for morphology) and interpreted lower expression and higher methylation of somitogenesis, mesoderm, gastrulation, BMP and Wnt genes in the IVP ED as an effect of the in vitro procedure on disc development. The present analysis reproduces the lower expression of these genes in same-length conceptuses and identifies an alternative explanation compatible with bulk expression data: much of the signal tracks the disc-enrichment score of each section. This does not establish that the regulatory interpretation of the parent study is incorrect, and the two explanations are not mutually exclusive. A testable hypothesis follows: if disc-specific genes are less methylated in the disc than in the membranes, a lower disc proportion in bulk ED samples would also appear as higher methylation.

Adjusting expression and methylation analyses of the full cohort for an independently validated measure of disc content would test this.

Unresolved questions and limitations. First, inferred sex is completely confounded with production group in the primary comparison, possibly because male embryos develop faster in vitro and more often reached the selected length. Sex determines the expression level of a large proportion of genes in bovine blastocysts (Bermejo-Álvarez et al., 2010); X-linked genes were over-represented among DEGs (8–10% vs 3% of tested genes), and although removing them did not change the gene-set results, effects of sex on autosomal genes cannot be excluded. Second, the four conceptuses of each group are not four independent maternal environments: the IVP conceptuses were carried by two recipients and the IVV conceptuses came from two superovulated donors, which also share maternal genetics within donor. Embryo-to-cow assignments were not available, the models do not account for embryos sharing a cow, and superovulation itself increases progesterone and can accelerate elongation (Clemente et al., 2009). This is a sampling limitation that model settings cannot remove. Third, IVV conceptuses were conceived by AI rather than by in vitro fertilisation with the same semen preparation (Behrens et al., 2025). Fourth, the four IVP conceptuses carry a sample-sheet label ("Vitrified") whose meaning is unresolved and which is associated with expression variation that cannot be separated from the group contrast. Fifth, conceptuses were selected by length category and presence of a disc, without measured lengths or a stated matching tolerance, so the results apply to a selected subset of surviving conceptuses. Sixth, the disc-enrichment score has not been validated against histology, disc area or independent cell-type references. Future work should use sex-balanced designs with more dams and known cow assignments, measure disc size, and apply single-cell or spatial transcriptomics to resolve differences in each cell lineage.

Conclusion

Selected IVP and IVV day-15 conceptuses of the same length differed in ED and EEM expression profiles. Glycosylation and lipid-metabolism gene sets were the most consistent ED-associated patterns across the analyses examined, and adhesion-associated gene sets were directionally more consistent in the EEM than translation and respiration gene sets. Developmental expression contrasts in the ED were sensitive to adjustment for an internally derived disc-enrichment score, leaving tissue composition and cell-intrinsic regulation unresolved. Because production group is completely confounded with inferred sex in the primary comparison, maternal replication is limited and part of the sample metadata is unresolved, these differences generate hypotheses but do not isolate effects of in vitro production.

Materials and methods

The analysis workflow, from read counts to differential expression, enrichment and the sensitivity analyses, is summarised in Supplementary Figure S1.

Animals and samples

Samples came from the study described by Behrens et al. (2025). For IVP, cumulus–oocyte complexes from slaughterhouse ovaries were matured, fertilised with frozen–thawed semen of a single Holstein bull and cultured in serum-free medium until day 7. Fresh IVP blastocysts were transferred non-surgically into two synchronised recipients. IVV embryos were produced by superovulation and AI of two donors with semen from the same bull. Conceptuses were flushed on day 15 and checked for the presence of an embryonic disc. Each selected embryo was cut into three pieces, the ED and two pieces of surrounding EEM, which were snap-frozen (Behrens et al., 2025). Libraries were sequenced paired-end, aligned to the ARS-UCD1.3 bovine genome and summarised as gene-level read counts. For this study, only conceptuses of the same length (sample-sheet category “elongated, short”) were used: four IVP and four IVV conceptuses, each contributing one ED and one EEM library. Measured lengths, the matching tolerance and embryo-to-cow assignments were not available. The IVP disc library IVPF10_EC, which has no matching EEM library, was excluded from the primary analysis.

Quality control, embryo sex and disc-specific transcript score

Library sizes, sample-to-sample correlations and PCA (Pearson, 1901; 500 most variable genes) were inspected. Embryo sex was inferred from XIST expression (< 1 vs > 25 counts per million; no library fell between these values). To relate the main axes of variation to sample characteristics, principal components were computed on the 500 most variable genes (log2 DESeq2-normalised counts) for all 32 libraries and separately for ED and EEM libraries, and the scores of the first ten components were correlated (Pearson) with tissue, production system, inferred sex, the unresolved sample-sheet label, membership of the same-length set, section, library size and, for ED libraries, the disc-specific transcript score; binary variables were coded 0/1. These correlations are descriptive and were not corrected for multiple testing. As an expression-based indicator of disc content, ED-enriched marker genes were defined as genes with log2 fold change > 1.5 in ED versus EEM in each of the four IVV conceptuses of the primary comparison and mean VST expression > 6 (263 genes); the disc-specific transcript score of a sample was the mean VST expression of these genes. Two expression transformations were used throughout: blind VST of all 32 libraries (DESeq2 vst, blind = TRUE) for the disc score and the group-separability analysis, and log2(DESeq2-normalised count + 1) for PCA (Figure 1, Supplementary Figure S3) and gene-level plots.

Differential expression

Analyses were performed in R 4.4.1 (R Core Team, 2023) with DESeq2 1.44.0 (Love et al., 2014). Genes with at least 10 reads in at least four libraries were retained. Differential expression was always tested on raw read counts with DESeq2 (negative binomial model, Wald test). IVP was compared with IVV in each tissue with the design ~ type (primary analysis), and, for the ED, additionally with ~ disc-specific transcript score + type (score centred and scaled). The score-adjusted model is an exploratory sensitivity analysis conditional on an expression-derived covariate and may remove both composition-associated and biologically regulated differences, so both models are reported and interpreted together. Log2 fold changes were shrunk with apeglm (Zhu et al., 2019) for ranking and plotting. DEGs were defined as genes with Benjamini–Hochberg adjusted p-value (FDR) < 0.05 (Benjamini & Hochberg, 1995). Further models excluded the EEM library IVP9_T1A or included IVPF10_EC. Two within-cohort sensitivity analyses used additional embryos from the same study: a sex-matched comparison of male embryos only (~ type; ED 7 IVP vs 2 IVV, EEM 5 vs 2), and a full-cohort model including embryos of all lengths (~ sex + length group + type, with length group coding same-length vs other embryos; ED also with the disc-specific transcript score). Embryo IVPF13 was excluded from both because its ED and EEM libraries disagreed on inferred sex (ED male, EEM female; XIST 0.1

vs 61 counts per million), which raises a sample-identity or sex-classification concern that cannot be resolved from the available metadata. Agreement with the primary analysis was assessed by the direction of effect of primary DEGs, the correlation of fold changes and GSEA of key terms. To assess the contribution of embryo sex directly, genes were mapped to chromosomes with org.Bt.eg.db, over-representation of X-linked genes among DEGs was tested with Fisher's exact test, and GSEA of key terms was repeated after removing all X-linked genes.

Comparison of tissues

To compare the IVP–IVV contrast between tissues, all 16 libraries were analysed together with the DESeq2 design ~ type + type:embryo_n + type:region, where embryo_n codes the four embryos 1–4 within each production type (the DESeq2 design for individuals nested within groups; 10 coefficients, 6 residual degrees of freedom). The difference between the group contrast in EEM and in ED was tested with results(dds, contrast = list("typeIVP.regionTE", "typeIVV.regionTE")) (Wald test, FDR < 0.05; in the code the EEM is labelled TE), so positive values mean a more positive IVP–IVV contrast in the EEM. This model was not adjusted for the disc-specific transcript score. Per-gene effect sizes were compared using apeglm-shrunken log2 fold changes on the genes tested in both tissues.

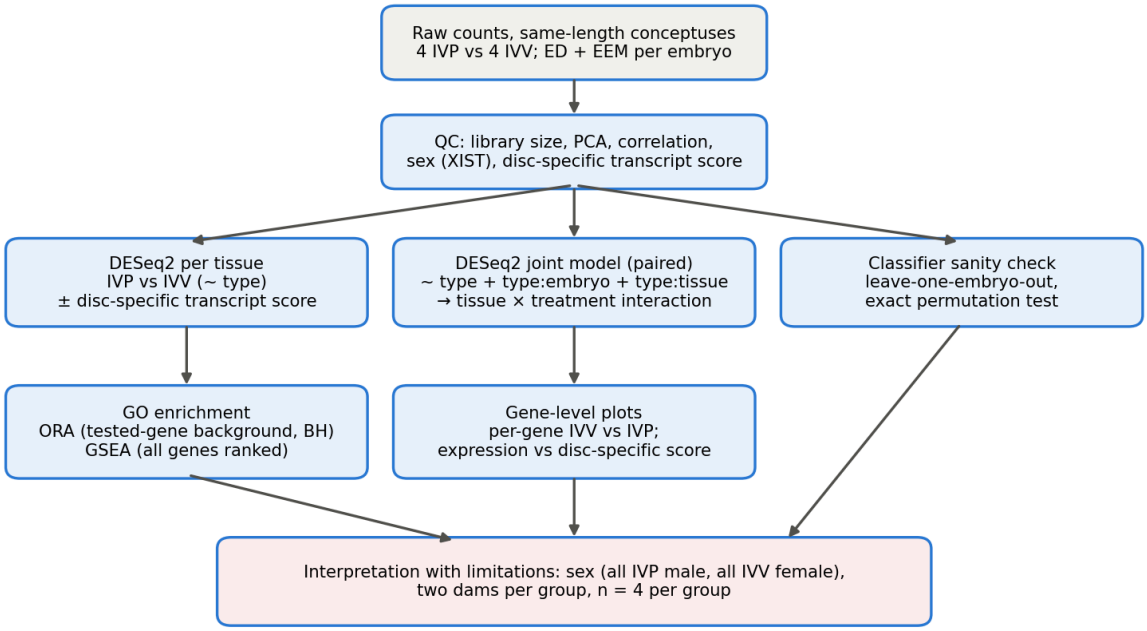
Functional enrichment

ORA of up- and down-regulated DEGs was performed with clusterProfiler (Wu et al., 2021) against Gene Ontology biological processes (Ashburner et al., 2000; Gene Ontology Consortium, 2023), using all genes tested in that tissue as background, BH correction, and term sizes of 10–500 genes; redundant terms were collapsed with simplify. GSEA (Subramanian et al., 2005) was performed with gseGO (GO biological processes) and gseKEGG (KEGG pathways, organism code bta; Kanehisa & Goto, 2000), both using the fgsea algorithm (Korotkevich et al., 2021), on all tested genes ranked by the DESeq2 Wald statistic, with gene-set sizes of 10–500 and BH FDR < 0.05. For the tissue × group interaction, genes were ranked by the Wald statistic of the interaction contrast, and only GSEA was performed. Gene annotation used org.Bt.eg.db. Genes for the signalling box plots were the ED DEGs in the leading edge (core enrichment) of the KEGG Wnt, TGF-β and Notch pathways in the ED GSEA. For the leading-edge analysis (Table 2), themes, representative sets and related sets were fixed before inspection; overlap between leading edges was summarised as the median pairwise Jaccard index, and sign agreement of leading-edge fold changes was compared with the score-adjusted, male-only and full-cohort analyses.

Sensitivity analyses on individual embryos and on the disc score

For a list of gene sets and genes fixed in advance (the main themes of this report), three further analyses were run. (1) Embryo influence: each of the eight primary conceptuses was omitted in turn (both of its libraries), and the ED model, the score-adjusted ED model (original marker set, score re-scaled), the EEM model and the tissue × group interaction model were refitted; for the interaction with three vs four conceptuses, the model matrix was passed to DESeq2 with empty columns removed. (2) EEM library IVP9_T1A was omitted. (3) Disc-marker selection: the marker set was rebuilt four times, each time omitting one of the four IVV conceptuses from the definition, and the score-adjusted ED model was refitted. Gene-set statistics for these analyses were computed with fgsea on the fixed GO terms only (gene sets from org.Bt.eg.db), ranked by the DESeq2 Wald statistic, and the reference values were recomputed the same way so that all columns are comparable. Direction of the group contrast was the primary outcome; FDR values are secondary, because each omission reduces the comparison to three versus four conceptuses.

Supplementary section

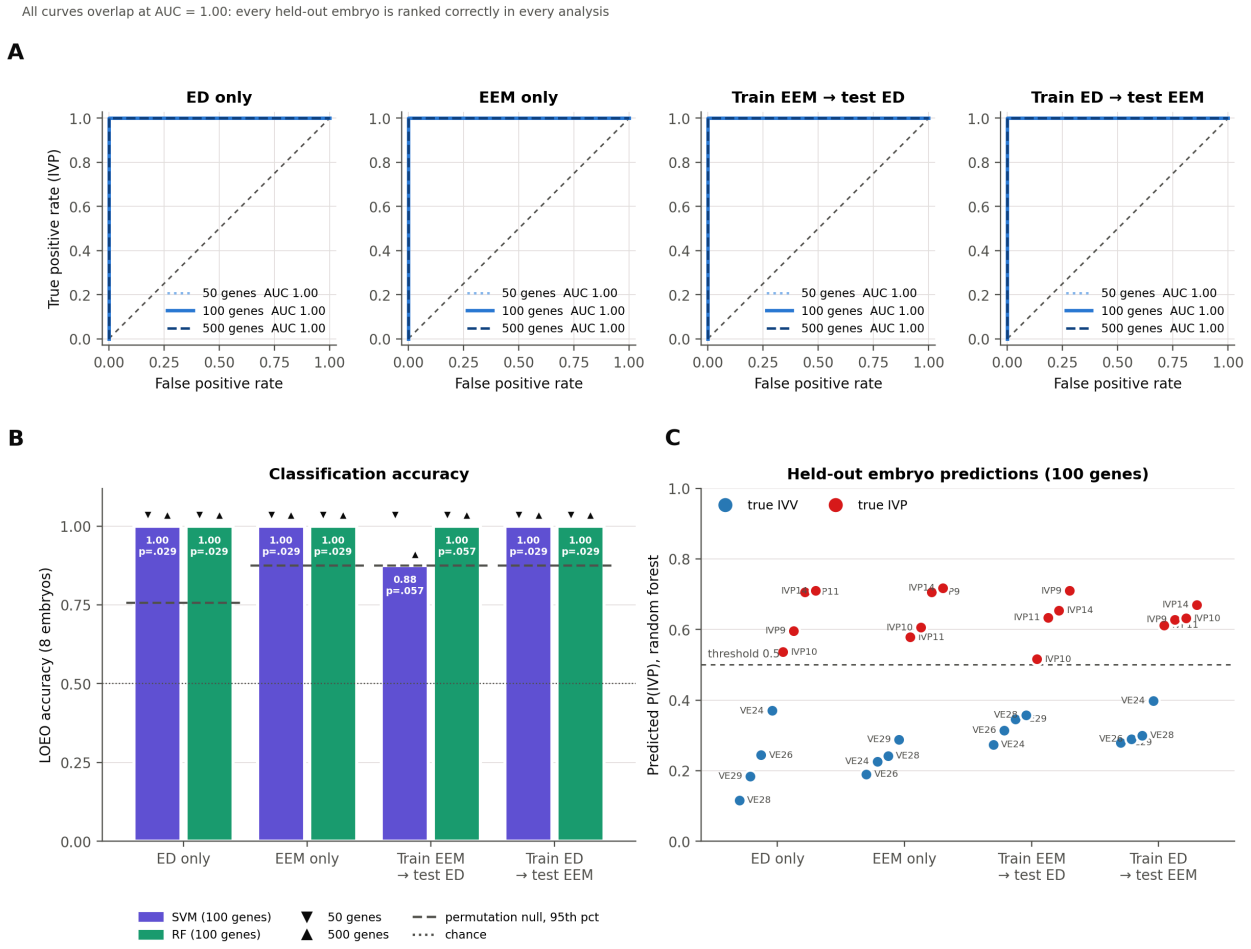


Supplementary Figure S1. Bioinformatic workflow.

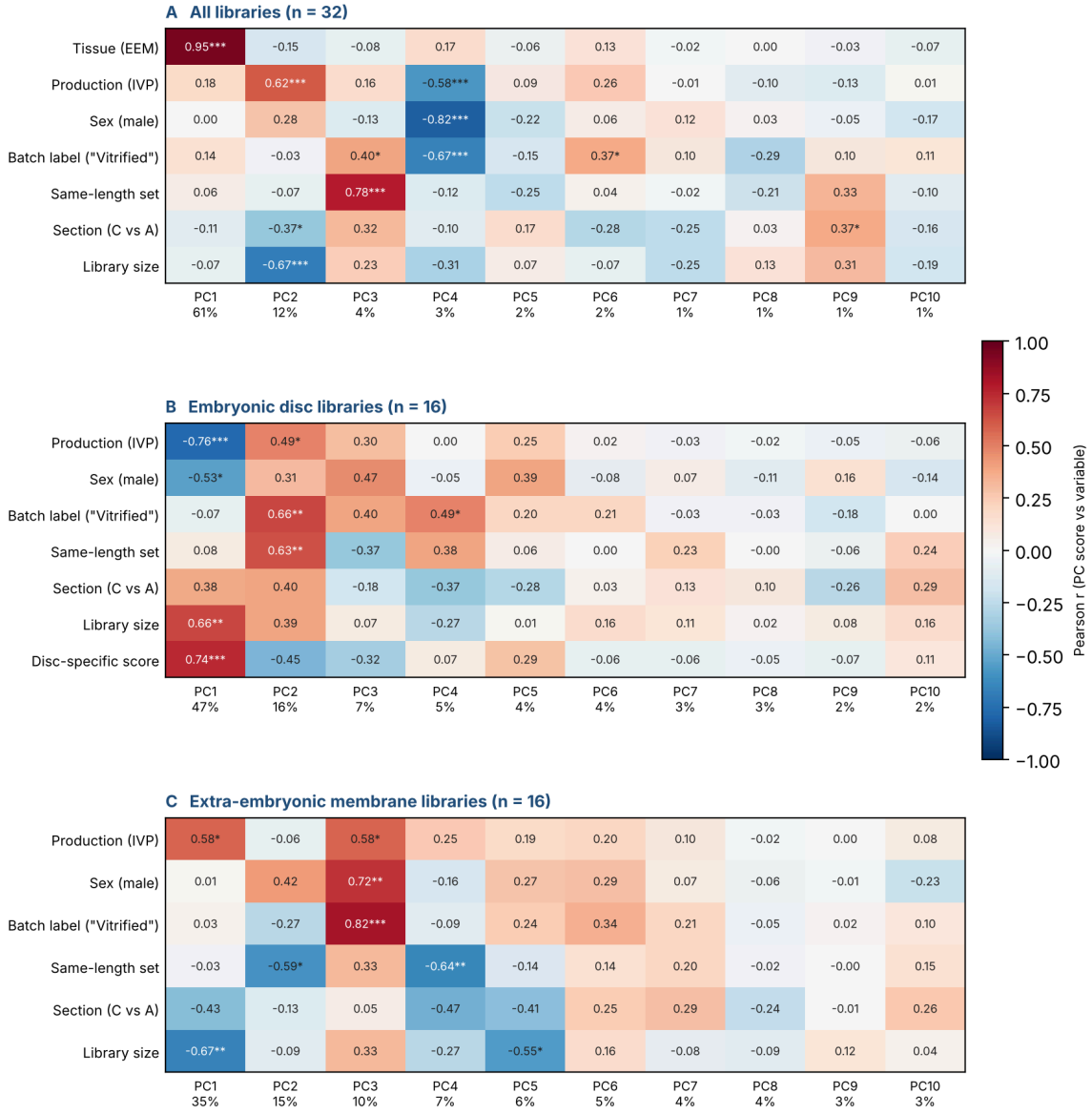
Supplementary analysis: exploratory group separability

To illustrate how strongly the observed groups separate, IVP and IVV conceptuses were classified by leave-one-embryo-out cross-validation from either tissue. Both linear SVM and random forest separated the groups perfectly from the ED or the EEM alone (accuracy 1.00; permutation p = 0.029 over the 70 embryo-level label assignments, the lowest value obtained), and a classifier trained on one tissue also classified the other (accuracy 0.875–1.00; Supplementary Figure S2). Perfect separation is expected given the many DEGs and does not show whether it reflects production system, sex, cow or other aligned factors. The permutation result assumes that embryos are exchangeable, which is questionable when conceptuses share a cow, and the genes used are not biomarkers.

Methods. Linear SVM (e1071, cost = 1; Meyer et al., 2023) and random forest (randomForest, 500 trees; Liaw & Wiener, 2002) were trained on blind-VST expression. In leave-one-embryo-out cross-validation, both libraries of the held-out embryo were removed from training, and the top 100 genes by Welch t-statistic were selected on the training embryos only. Exact permutation p-values were obtained by repeating the full procedure for all 70 embryo-level label assignments. For cross-tissue prediction, each gene was z-scored within tissue and a model trained on one tissue was applied to the other tissue of the held-out embryo. Label-driven gene selection was restricted to training embryos, but the unsupervised transformations (blind VST and within-tissue standardisation) used all samples, including the held-out embryo; the gene numbers 50, 100 and 500 were fixed in advance and all are reported.



Supplementary Figure S2. Exploratory group-separability analysis. (A) ROC curves (random forest) for within-tissue and cross-tissue prediction at 50, 100 and 500 genes; all AUC = 1.00. (B) Leave-one-embryo-out accuracy (100 genes; triangles, 50 and 500 genes); dashed line, 95th percentile of the permutation null; p, exact permutation p-value over 70 embryo-level labellings. (C) Predicted probability of IVP for each held-out embryo.



Supplementary Figure S3. Correlation of principal components (PCs) with sample metadata for (A) all 32 libraries, (B) ED libraries and (C) EEM libraries. PCs were computed from the 500 most variable genes (log2 DESeq2-normalised counts); PC1 captures the largest source of variation between samples, PC2 the next, and so on, and the percentage under each PC is the share of variance it explains. Each cell gives the Pearson correlation (r, −1 to +1) between the samples' scores on that PC and the variable in that row; values near ±1 mean that the PC separates the groups almost perfectly, values near 0 mean no relationship. Two-group variables are coded 0/1, with the group named in brackets coded 1 (e.g. Tissue (EEM): EEM = 1, ED = 0). Red, the group coded 1 scores higher on the PC; blue, lower; the sign of a PC is arbitrary, so the strength of the correlation is what matters. Library size, millions of assigned reads; disc-specific score, mean expression of disc marker genes (ED panel only). * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ (uncorrected; correlations on PCs that explain little variance should be interpreted cautiously). Variables that do not vary within a panel are omitted.

Supplementary Table S1. All libraries of the cohort and their use in each analysis

XIST, counts per million; sex inferred from XIST (< 1 male, > 25 female). Disc score, disc-specific transcript score (same-length ED libraries only). Sample-sheet label, recorded value of the sample-sheet field “cryopreserved”; its meaning is unresolved, and all IVP embryos were transferred fresh (Behrens et al., 2025). Primary, primary comparison (8 ED + 8 EEM libraries); male-only and full cohort, within-cohort sensitivity analyses. Conceptus lengths and embryo-to-cow assignments: not available. IVP10 and IVPF10 are different conceptuses.

Library	Embryo	Group	Tissue	XIST	Inferred sex	Disc score	Sample-sheet label	Length category	Primary	Male-only	Full cohort	Note
VE24_EC	VE24	IVV	ED	53.5	female	8.32	Fresh	same-length	✓	–	✓	
VE24_T1C	VE24	IVV	EEM	43.9	female	–	Fresh	same-length	✓	–	✓	
VE26_EC	VE26	IVV	ED	47.9	female	8.88	Fresh	same-length	✓	–	✓	
VE26_T1C	VE26	IVV	EEM	33.6	female	–	Fresh	same-length	✓	–	✓	
VE28_EC	VE28	IVV	ED	52.7	female	9.13	Fresh	same-length	✓	–	✓	
VE28_T1C	VE28	IVV	EEM	37.0	female	–	Fresh	same-length	✓	–	✓	
VE29_EC	VE29	IVV	ED	56.8	female	9.27	Fresh	same-length	✓	–	✓	
VE29_T1C	VE29	IVV	EEM	28.9	female	–	Fresh	same-length	✓	–	✓	
IVP10_EC	IVP10	IVP	ED	0.3	male	7.33	Vitrified	same-length	✓	✓	✓	
IVP10_T1C	IVP10	IVP	EEM	0.1	male	–	Vitrified	same-length	✓	✓	✓	
IVP11_EC	IVP11	IVP	ED	0.1	male	7.73	Vitrified	same-length	✓	✓	✓	
IVP11_T1C	IVP11	IVP	EEM	0.0	male	–	Vitrified	same-length	✓	✓	✓	
IVP14_EC	IVP14	IVP	ED	0.0	male	8.47	Vitrified	same-length	✓	✓	✓	
IVP14_T1C	IVP14	IVP	EEM	0.1	male	–	Vitrified	same-length	✓	✓	✓	
IVP9_EA	IVP9	IVP	ED	0.0	male	8.47	Vitrified	same-length	✓	✓	✓	
IVP9_T1A	IVP9	IVP	EEM	0.6	male	–	Vitrified	same-length	✓	✓	✓	Lowest within-group correlation
IVPF10_EC	IVPF10	IVP	ED	0.0	male	7.77	Fresh	same-length	–	✓	✓	No matching EEM library; not in primary comparison
VE19_EA	VE19	IVV	ED	104.7	female	–	Fresh	other	–	–	✓	
VE19_T1A	VE19	IVV	EEM	67.7	female	–	Fresh	other	–	–	✓	
VE3_EA	VE3	IVV	ED	0.1	male	–	Fresh	other	–	✓	✓	

VE3_T1A	VE3	IVV	EEM	0.1	male	–	Fresh	other	–	✓	✓	
VE4_EC	VE4	IVV	ED	77.6	female	–	Fresh	other	–	–	✓	
VE4_T1C	VE4	IVV	EEM	40.9	female	–	Fresh	other	–	–	✓	
VE5_EC	VE5	IVV	ED	0.1	male	–	Fresh	other	–	✓	✓	
VE5_T1C	VE5	IVV	EEM	0.1	male	–	Fresh	other	–	✓	✓	
VE6_EC	VE6	IVV	ED	130.3	female	–	Fresh	other	–	–	✓	
IVPF13_EA	IVPF13	IVP	ED	0.1	male	–	Fresh	other	–	–	–	Tissues disagree on inferred sex; excluded
IVPF13_T1A	IVPF13	IVP	EEM	61.4	female	–	Fresh	other	–	–	–	Tissues disagree on inferred sex; excluded
IVPF14_EA	IVPF14	IVP	ED	0.2	male	–	Fresh	other	–	✓	✓	
IVPF14_TA	IVPF14	IVP	EEM	0.1	male	–	Fresh	other	–	✓	✓	
IVPF15_EC	IVPF15	IVP	ED	0.1	male	–	Fresh	other	–	✓	✓	
IVPF9_T1A	IVPF9	IVP	EEM	102.6	female	–	Fresh	other	–	–	✓	

Supplementary Table S2. Strongest GSEA terms per analysis

Top eight GO biological-process terms and top five KEGG pathways in each direction (lowest FDR) for IVP vs IVV in the ED and EEM and for the tissue × group interaction. Size, number of tested genes in the set.

Analysis	Database	Direction	Term	Size	NES	FDR
ED	GO BP	higher in IVP	cellular lipid metabolic process	408	+1.80	8.7e-06
ED	GO BP	higher in IVP	glycosylation	129	+2.15	1.5e-05
ED	GO BP	higher in IVP	protein glycosylation	122	+2.14	2.4e-05
ED	GO BP	higher in IVP	macromolecule glycosylation	122	+2.14	2.4e-05
ED	GO BP	higher in IVP	lipid biosynthetic process	247	+1.92	2.4e-05
ED	GO BP	higher in IVP	glycoprotein biosynthetic process	151	+2.01	4.0e-05
ED	GO BP	higher in IVP	glycoprotein metabolic process	179	+1.92	2.4e-04
ED	GO BP	higher in IVP	glycerolipid metabolic process	165	+1.92	2.4e-04
ED	GO BP	lower in IVP	anatomical structure morphogenesis	408	-1.93	2.4e-07
ED	GO BP	lower in IVP	nervous system development	452	-1.69	2.4e-04
ED	GO BP	lower in IVP	embryo development	73	-2.05	5.5e-04
ED	GO BP	lower in IVP	system process	161	-1.86	6.1e-04
ED	GO BP	lower in IVP	neuron differentiation	276	-1.72	7.7e-04
ED	GO BP	lower in IVP	cell-cell signaling	333	-1.67	8.9e-04
ED	GO BP	lower in IVP	animal organ development	262	-1.73	9.4e-04
ED	GO BP	lower in IVP	neurogenesis	319	-1.65	1.4e-03
ED	KEGG	higher in IVP	Glycerophospholipid metabolism	88	+1.97	4.1e-04
ED	KEGG	higher in IVP	Glycerolipid metabolism	56	+1.96	3.4e-03
ED	KEGG	higher in IVP	Glycosphingolipid biosynthesis - lacto and neolacto series	23	+2.09	3.9e-03
ED	KEGG	higher in IVP	Sphingolipid metabolism	50	+1.88	7.4e-03
ED	KEGG	higher in IVP	Ether lipid metabolism	38	+1.95	8.3e-03
ED	KEGG	lower in IVP	Transcriptional misregulation in cancer	146	-2.21	6.7e-08

ED	KEGG	lower in IVP	Basal cell carcinoma	58	-2.19	1.7e-05
ED	KEGG	lower in IVP	Systemic lupus erythematosus	64	-2.19	2.2e-05
ED	KEGG	lower in IVP	Neutrophil extracellular trap formation	115	-1.99	8.3e-05
ED	KEGG	lower in IVP	Alcoholism	132	-2.00	1.1e-04
EEM	GO BP	higher in IVP	cell adhesion	252	+1.94	9.2e-06
EEM	GO BP	higher in IVP	axon development	106	+2.06	5.8e-05
EEM	GO BP	higher in IVP	axonogenesis	100	+2.06	7.9e-05
EEM	GO BP	higher in IVP	axon guidance	70	+2.10	2.7e-04
EEM	GO BP	higher in IVP	neuron projection guidance	70	+2.10	2.7e-04
EEM	GO BP	higher in IVP	regulation of biological quality	418	+1.64	3.0e-04
EEM	GO BP	higher in IVP	cell migration	182	+1.85	3.4e-04
EEM	GO BP	higher in IVP	nervous system development	381	+1.69	3.4e-04
EEM	GO BP	lower in IVP	translation	319	-2.11	9.1e-10
EEM	GO BP	lower in IVP	peptide biosynthetic process	325	-2.07	1.6e-09
EEM	GO BP	lower in IVP	ribonucleoprotein complex biogenesis	265	-1.97	5.4e-07
EEM	GO BP	lower in IVP	amide biosynthetic process	368	-1.88	5.4e-07
EEM	GO BP	lower in IVP	protein folding	126	-2.21	3.9e-06
EEM	GO BP	lower in IVP	cellular respiration	80	-2.32	5.1e-06
EEM	GO BP	lower in IVP	ribosome biogenesis	202	-1.96	6.1e-06
EEM	GO BP	lower in IVP	aerobic respiration	73	-2.33	1.2e-05
EEM	KEGG	higher in IVP	Cytoskeleton in muscle cells	162	+2.21	6.0e-08
EEM	KEGG	higher in IVP	Integrin signaling	113	+2.09	1.9e-05

EEM	KEGG	higher in IVP	Calcium signaling pathway	156	+1.84	5.3e-04
EEM	KEGG	higher in IVP	Cytokine-cytokine receptor interaction	121	+1.83	5.4e-04
EEM	KEGG	higher in IVP	ECM-receptor interaction	59	+2.04	5.9e-04
EEM	KEGG	lower in IVP	Protein processing in endoplasmic reticulum	162	-2.48	6.0e-11
EEM	KEGG	lower in IVP	Ribosome	180	-2.35	2.0e-10
EEM	KEGG	lower in IVP	Systemic lupus erythematosus	56	-2.57	6.0e-08
EEM	KEGG	lower in IVP	Viral carcinogenesis	162	-2.04	5.3e-06
EEM	KEGG	lower in IVP	Neutrophil extracellular trap formation	107	-2.16	1.2e-05
Interaction (EEM − ED)	GO BP	more positive in EEM	nervous system development	461	+2.07	3.4e-12
Interaction (EEM − ED)	GO BP	more positive in EEM	anatomical structure morphogenesis	425	+2.07	8.4e-12
Interaction (EEM − ED)	GO BP	more positive in EEM	cell adhesion	317	+2.14	1.0e-10
Interaction (EEM − ED)	GO BP	more positive in EEM	neurogenesis	326	+2.06	1.9e-10
Interaction (EEM − ED)	GO BP	more positive in EEM	generation of neurons	297	+2.09	1.6e-09
Interaction (EEM − ED)	GO BP	more positive in EEM	neuron differentiation	283	+2.09	2.5e-09
Interaction (EEM − ED)	GO BP	more positive in EEM	cell morphogenesis	208	+2.11	8.2e-09
Interaction (EEM − ED)	GO BP	more positive in EEM	neuron development	226	+2.11	2.8e-08
Interaction (EEM − ED)	GO BP	more negative in EEM	establishment of protein localization	453	-2.38	1.4e-19
Interaction (EEM − ED)	GO BP	more negative in EEM	protein transport	394	-2.37	7.2e-18
Interaction (EEM − ED)	GO BP	more negative in EEM	intracellular protein transport	291	-2.44	2.3e-15
Interaction (EEM − ED)	GO BP	more negative in EEM	Golgi vesicle transport	166	-2.63	4.0e-12
Interaction (EEM − ED)	GO BP	more negative in EEM	endoplasmic reticulum to Golgi vesicle-mediated transport	85	-2.86	6.8e-12
Interaction (EEM − ED)	GO BP	more negative in EEM	protein folding	129	-2.44	3.1e-09

Interaction (EEM − ED)	GO BP	more negative in EEM	ribonucleoprotein complex biogenesis	270	-2.12	3.5e-09
Interaction (EEM − ED)	GO BP	more negative in EEM	establishment of protein localization to organelle	168	-2.34	5.9e-09
Interaction (EEM − ED)	KEGG	more positive in EEM	Cytoskeleton in muscle cells	190	+2.20	5.2e-09
Interaction (EEM − ED)	KEGG	more positive in EEM	Cell adhesion molecule (CAM) interaction	106	+2.19	7.0e-07
Interaction (EEM − ED)	KEGG	more positive in EEM	ECM-receptor interaction	73	+2.22	6.8e-06
Interaction (EEM − ED)	KEGG	more positive in EEM	Axon guidance	163	+2.00	7.8e-06
Interaction (EEM − ED)	KEGG	more positive in EEM	Integrin signaling	131	+1.96	1.8e-05
Interaction (EEM − ED)	KEGG	more negative in EEM	Protein processing in endoplasmic reticulum	164	-3.06	1.2e-21
Interaction (EEM − ED)	KEGG	more negative in EEM	Endocytosis	230	-2.01	1.2e-07
Interaction (EEM − ED)	KEGG	more negative in EEM	Nucleocytoplasmic transport	103	-2.26	1.1e-06
Interaction (EEM − ED)	KEGG	more negative in EEM	N-Glycan biosynthesis	56	-2.29	1.8e-05
Interaction (EEM − ED)	KEGG	more negative in EEM	Lysosome biogenesis	244	-1.74	5.8e-05

Data and code availability

Raw sequencing data are available in GEO under accession GSE278243 (Behrens et al., 2025). All analysis code, the sample sheet, result tables and figures are available in the accompanying GitHub repository (see its README for how to obtain the gene-level count matrix and reproduce every result).

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